Effect of Exercise on Cognition, Memory, and Executive Function: A Study-Level Meta-Meta-Analysis Across Populations and Exercise Categories

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Abstract

Background: Physical exercise is widely promoted for cognitive enhancement, yet prior syntheses have been questioned due to methodological limitations.

Objectives: This study evaluated the reliability of evidence supporting physical exercise for general cognition, memory, and executive function, and assessed efficacy across specific population-intervention combinations after adjusting for publication bias.

Methods: This study is a study-level meta-meta-analysis of randomized controlled trials (RCTs). We included systematic reviews with meta-analyses of RCTs evaluating exercise interventions against non-exercise controls in humans, identified in a meta-meta-analysis published in 2025. We extracted study-level effect sizes and synthesized the data using robust Bayesian meta-analysis which allowed us to evaluate evidence for and against the presence of the effect, between-study heterogeneity, publication bias, while accounting for model uncertainty.

Results: We extracted 2,239 study-level effect estimates from 215 meta-analyses. After excluding suspected duplicates, 1,933 estimates from 201 meta-analyses were included in the overall analyses, which revealed extreme evidence of publication bias and between-study heterogeneity. Adjusted average effects were significantly smaller than unadjusted estimates for general cognition (SMD=0.208; 95% credible interval (CrI) [0.075, 0.318]), and executive functions (SMD=0.088, 95%CrI [0.000, 0.168]). For memory the adjustment differed between adjustment specifications (SMD=0.226, 95%CrI [0.169, 0.282] vs SMD=0.021, 95%CrI [0.000, 0.217]). While broad prediction intervals spanned substantial negative and positive effects, subgroup analyses identified mostly consistent benefits for specific combinations, such as dance and exergaming for general cognition in non-healthy older adults.

Conclusion: The primary limitations of the evidence are severe publication bias and extreme heterogeneity, which appear to inflate previously reported benefits. Consequently, broad claims of generalized cognitive enhancement from physical exercise appear

premature. The results support a shift from blanket public health advice to recommendations tailored to specific population-intervention combinations.

Effect of Exercise on Cognition, Memory, and Executive Function: A Study-Level Meta-Meta-Analysis Across Populations and Exercise Categories

Introduction

There are well-documented benefits of physical exercise for physiological and health outcomes (Sallis et al., 2016). In addition, the more recent accumulation of studies showing a positive effect of physical exercise on cognitive function is reflected in the updated physical activity guidelines (Ding et al., 2020). Accordingly, major public health agencies promote exercise for brain health, including the Centers for Disease Control and Prevention (Piercy et al., 2018) and the World Health Organization (Bull et al., 2020).

A critical umbrella review of randomized controlled trials (RCTs) by Ciria et al. (2023) re-examined 24 meta-analyses and highlighted serious issues: underpowered designs, salami slicing, regression to the mean, placebo-like effects, selective reporting, and publication bias. Their re-analysis found small average benefits (standardized mean difference, SMD = 0.22) that attenuated after accounting for key moderators (active control groups and baseline differences; SMD = 0.13) and became negligible after publication-bias adjustment (SMD = 0.05). See also the subsequent discussion focused on inclusion criteria, populations, moderators, and causal mechanisms (Ciria et al., 2024; Dupuy et al., 2024).

A more recent umbrella review and meta-meta-analysis provided a comprehensive synthesis of 133 systematic reviews covering 2,724 RCTs and 258,279 participants (Singh et al., 2025). Pooling meta-analysis-level effects, it reported a substantial effect of exercise on general cognition (SMD = 0.42) and moderate effects on memory (SMD = 0.26) and executive function (SMD = 0.24). They concluded that "these findings provide strong evidence that exercise, even light intensity, benefits general cognition, memory and executive function across all populations, reinforcing exercise as an essential, inclusive recommendation for optimising cognitive health" Singh et al., 2025, abstract. By design, this careful meta-meta-analysis aggregated meta-analysis-level estimates rather than study-level data. Consequently, it could not fully adjust for publication bias or quantify

between-study heterogeneity at the study level.

To address these limitations, we compile a dataset of study-level effect sizes and extract key design characteristics from the RCTs identified by Singh et al. (2025). This granular approach allows us to adjust for publication bias, accurately quantify between-study heterogeneity, and conduct detailed subgroup analyses across populations and exercise types.

This study advances the exercise–cognition literature by providing a comprehensive, study-level perspective that earlier meta-meta-analyses and umbrella reviews could not offer. First, by assembling a large, study-level dataset across populations and exercise types, we provide an updated and empirically grounded map of the conditions under which cognitive benefits are most likely to occur. This granularity allows us to move beyond the assumption of a generalized effect. Second, by combining study-level re-analysis with advanced meta-analytic methodology that explicitly accounts for model uncertainty, between-study heterogeneity, and publication bias, we ensure the robustness of our results. Taken together, these features yield a nuanced evidence base to inform realistic expectations and targeted recommendations regarding for whom, and under what conditions, exercise is likely to improve cognition.

Methods

Literature search, inclusion, and exclusion criteria

This project extends the meta-meta-analysis by Singh et al. (2025) and therefore relies on its search, screening, and inclusion procedures. The meta-meta-analysis's inclusion criteria were based on the following population, intervention, comparison, outcomes, and study type (PICOS) criteria (adopted from page 2 of Singh et al., 2025):

- Population: any human population (children, adolescents, adults; healthy and clinical).

- Intervention: Reviews that evaluated exercise interventions were included. The following definition of exercise was used: 'a type of physical activity consisting of planned, structured and repetitive bodily movement done to improve and/or maintain physical fitness'. Reviews were included if $\geq 75\%$ of the included RCTs focused solely on exercise, including (but not limited to) aerobic or resistance exercise, yoga, dance, Tai Chi and exergames, which were not combined with any other intervention. Reviews evaluating regular exercise training of at least 4 weeks were included irrespective of exercise mode, supervision, delivery, intensity or weekly duration.

- Comparator: reviews were eligible if $\geq 75\%$ of the included RCTs compared exercise to no intervention, waitlist, usual care, nothing, a sham intervention, an equal attention non-exercise intervention arm or a lower/lesser exercise intervention.

- Outcomes: any assessment of general cognition, memory or executive function.

- Study type: systematic reviews that included meta-analyses. Reviews were excluded if they included any nonRCTs or studies assessing single bouts of exercise.

As described in Singh et al. (2025), the included databases were "CINAHL, The Cochrane Library, Embase via OVID, MEDLINE via OVID, Emcare via OVID, ProQuest Central, ProQuest Nursing and Allied Health Source, PsycINFO, Scopus, Sport Discus via Ebscohost and Web of Science using subject heading, keyword and MeSH term searches for 'systematic review', 'meta-analysis', 'cognitive function', 'memory', 'executive function', and 'exercise"' (p. 2). The original database search was limited to peer-reviewed journal articles published in English-language up to 1 November 2023. See Online Supplements of Singh et al. (2025) search strategy details. See Figure 1 in Singh et al. (2025) for PRISMA flowchart detailing the search, inclusion, and exclusion criteria resulting in 133 included studies.

Data extraction

We manually extracted study-level effect sizes and standard errors (or other information required for their computation, such as confidence intervals or summary statistics) from all meta-analyses (from study information tables, forest plots, and online supplementary materials) included in eFigure 1, 2, and 3 in the Supplementary Materials of Singh et al. (2025). The necessary information was available for 215 out of the 286 meta-analyses (with 2,239 estimates) reported by Singh et al. (2025). Bayesian linear-probability models with default prior distributions (Rouder & Morey, 2012) found weak to moderate evidence against association between study-level data availability and the year of publication ($BF_{10} = 0.142$), the meta-analytic effect size ($BF_{10} = 0.199$) or the outcome category ($BF_{10} = 0.640$).

The correctness of the extracted study-level estimates was assessed by comparing the pooled meta-analytic estimates reported in Singh et al. (2025) to the pooled meta-analytic estimates re-computed from the study-level estimates (the closest pooled estimate from either a fixed-effect meta-analysis or a random-effects meta-analysis with restricted maximum likelihood, maximum likelihood, and the DerSimonian-Laird estimator implemented in the `metafor` `R` package (Viechtbauer, 2010)). The absolute difference between the reported and the closest re-computed pooled estimate was lower than SMD = 0.02 for 201 meta-analyses, between SMD = 0.02 and SMD = 0.05 for 7 meta-analyses, and larger than SMD = 0.05 for 7 meta-analyses. All meta-analyses with absolute differences larger than SMD = 0.05 were re-examined by a second coder and the discrepancies were attributed to typos/typesetting in the original meta-analyses and the data were retained. Two separate datasets were created from the data: a) a meta-analysis level re-analysis dataset with a total of 215 meta-analyses containing 2,239 effect size estimates and b) an overall re-analysis dataset containing 1,933 effect size estimates from 201 meta-analyses (306 effect size estimates were assessed to be duplicated across meta-analyses based on a three decimal match in effect sizes, standard errors, and study

characteristics described below).

In addition, we extracted information about the population type (healthy vs non-healthy), population age (children and adolescents, adults, and older adults), intervention duration (from days to years), exercise intensity (low and moderate or low, moderate, and high), and exercise types from eTable 3 from the Supplementary Materials of Singh et al. (2025). Since many meta-analyses included multiple exercise types, we attempted to extract the specific exercise types for each study-level estimate within each meta-analysis that involved multiple exercise types. The extracted exercise types were classified into the following categories: dance (109 estimates), exergaming (84 estimates), mind and body (exercises described as including tai-chi, yoga, baduanjin, etc., 292 estimates), resistance (exercises described as including 'resistance' or 'strength', 238 estimates), walking (79 estimates), and other (all estimates not matching any of the previous categories, 1131 estimates).

Statistical analysis

All analyses were performed on standardized mean differences (either Cohen's d or Hedge's g) as reported in the original meta-analyses. For 360 estimates, both effect sizes and standard errors were directly available. For 1,817 estimates, standard errors were computed using a normal approximation based on the confidence interval width. For 62 estimates, standard errors were computed from the effect size and total sample size under the assumption of equal group sizes Equation 4.20 Borenstein et al., 2009.

Analytical framework

Throughout the manuscript, we present results from three types of meta-analytic ensembles: publication bias-unadjusted Bayesian model-averaged meta-analysis (Bartoš et al., 2021; Berkhout et al., 2023; Gronau et al., 2021) and two versions of publication bias-adjusted robust Bayesian meta-analysis RoBMA, Bartoš et al., 2022; Maier et al., 2023. All three meta-analytic models employ Bayesian model averaging BMA, Fragoso

et al., 2018; Hoeting et al., 1999 to account for model uncertainty. In other words, instead of specifying a single meta-analytic model, BMA allowed us to specify a set of plausible meta-analytic models. The specified models are compared on the observed data based on their prior predictions; models that predicted the data well receive an increase in their posterior probability, while models that predicted the data poorly suffer a decline see e.g., Hinne et al., 2020, for an accessible introduction. Importantly, no model is completely excluded from the inference; as such, the model uncertainty is retained when interpreting the results see e.g., Wagenmakers et al., 2022.

Parameter estimates, e.g., the pooled effect (μ) or between-study heterogeneity (τ), are obtained by combining the posterior distributions across all models proportionally to their posterior model probabilities. As such, all reported estimates in the manuscript correspond to model-averaged posterior estimates [95% central credible intervals] as defined in Equation 11 in Bartoš et al. (2022). The main advantage of the model-averaged posterior estimates is that they incorporate uncertainty about the most appropriate data-generating model. Even if the true data-generating model is not part of the specified models, the model-averaged posterior distribution minimizes the Kullback–Leibler divergence to the true model (Kleijn & van der Vaart, 2006).

Hypothesis tests, e.g., the Bayes factor (Jeffreys, 1931; Kass & Raftery, 1995) for the presence of the effect (BF_{10}), heterogeneity ($BF_{\mathbf{rf}}$) or publication bias (BF_{bias}), are obtained by comparing the relative predictive performance of the set of models assuming the presence of the effect (or heterogeneity/publication bias) to the set of models assuming the absence of the effect (or heterogeneity/publication bias), i.e., the inclusion Bayes factor e.g., Hinne et al., 2020 as defined in Equation 10 in Bartoš et al. (2022). The main advantage of Bayes factors is that they allow us to directly interpret the strength of evidence and distinguish the evidence of absence from the absence of evidence e.g., Keysers et al., 2020. We follow the rules of thumb introduced in Appendix I of Jeffreys (1939) and on page 105 of Lee and Wagenmakers (2005) when interpreting Bayes factors.

Meta-analytic ensembles

The publication bias–unadjusted Bayesian model-averaged meta-analysis accounts for uncertainty about the presence or absence of an effect and heterogeneity. It therefore specifies four models: (1) effect present, heterogeneity present; (2) effect present, heterogeneity absent; (3) effect absent, heterogeneity present; and (4) effect absent, heterogeneity absent see e.g., Gronau et al., 2021.

The publication bias-adjusted RoBMA extends the publication bias-unadjusted model-averaged meta-analysis by also incorporating models that account for publication bias (Maier et al., 2023). We employ the $\text{RoBMA}_{\text{PSMA}}$ version (Bartoš et al., 2022) that combines two common and well-tested publication bias adjustment methods: selection models (Vevea & Hedges, 1995) and PET-PEESE (Stanley & Doucouliagos, 2014). While selection models directly adjust for selection for statistical significance or direction of the effect, PET-PEESE specifies meta-regressions with either a linear or quadratic standard error term to adjust for effect size inflation in smaller studies. In total, $\text{RoBMA}_{\text{PSMA}}$ specifies 8 publication bias adjustment models: 6 weight functions that allow for two- and one-sided selection on either statistically significant, marginally significant, or directionally aligned results, PET, and PEESE model see Bartoš et al., 2022, for details. These 8 models expand the set of previously described 4 models in Bayesian model-averaged meta-analysis, resulting in $4 \times 9 = 36$ separate models (8 publication bias adjustments and 1 publication bias unadjusted model type). See Bartoš et al. (2022) and Bartoš, Pawel, and Siepe (2025) for large-scale simulations collected by Hong and Reed (2020) demonstrating the performance of $\text{RoBMA}_{\text{PSMA}}$ across four independently developed simulation environments: Alinaghi and Reed (2018), Bom and Rachinger (2019), Carter et al. (2019), and Stanley et al. (2017) and empirical examples, including a demonstration of performance in the absence of publication bias by re-analyzing sets of registered replication reports.

As a sensitivity analysis, we present results from RoBMA using only the weight functions for publication bias adjustment (RoBMA_{WF}). Removing the PET-PEESE

publication bias adjustment part from $\text{RoBMA}_{\text{PSMA}}$ allows us to draw conclusions that are insensitive to potential systematic differences in populations and effect sizes. In other words, smaller studies might be conducted on clinical populations and result in a larger benefit, while larger studies might be conducted on the general population and result in a smaller benefit. This pattern of results might be inadvertently identified as small-study effects and consequently over-adjusted for by PET-PEESE (those patterns would not be identified as selection on statistical significance or direction of the effect by selection models). Removal of PET and PEESE models results in RoBMA_{WF} based on $4 \times 7 = 28$ separate models (6 publication bias adjustments and 1 publication bias unadjusted model type).

All three meta-analytic ensembles were extended to their meta-regression versions when testing for moderation and multilevel versions when dealing with estimates from different meta-analyses as described in Bartoš, Maier, Stanley, and Wagenmakers (2025) and Bartoš, Maier, and Wagenmakers (2025). All ensembles were estimated using the `RoBMA R` package version 3.6.1, Bartoš and Maier, 2020 in `R` (R Core Team, 2021).

Prior distributions

We specified default prior distributions as described in Bartoš et al. (2022); a standard normal prior distribution for the pooled effect size parameter, $\mu \sim \text{Normal}(0, 1)$, under the alternative hypothesis of the presence of an effect; an empirically informed inverse-gamma distribution for the between-study heterogeneity parameter, $\tau \sim \text{Inv-Gamma}(1, 0.15)$ (van Erp et al., 2017), under the alternative hypothesis of the presence of between-study heterogeneity; unit cumulative Dirichlet priors for the relative publication probabilities under the publication bias operating via selection models; a unit Cauchy prior distribution for the PET regression coefficient, $\beta_{\text{PET}} \sim \text{Cauchy}(0, 1)$, under the publication bias operating via the PET model; and a Cauchy prior distribution for the PEESE regression coefficient, $\beta_{\text{PEESE}} \sim \text{Cauchy}(0, 5)$, under the publication bias operating

via the PEESE model. The null hypotheses were specified via spikes at zero, i.e., $\mu = 0$ and $\tau = 0$.

Meta-regression used the default prior distributions as described in Bartoš, Maier, Stanley, and Wagenmakers (2025) on the grand mean-difference contrast coding, i.e., Normal$(0, 0.25)$ marginal prior distribution on the difference between each moderator level and the grand mean μ. Multilevel meta-analysis used the default uniform prior distribution, as described in Bartoš, Maier, Stanley, and Wagenmakers (2025), on the within/between allocation via a heterogeneity allocation parameter $\rho \sim \text{Uniform}(0, 1)$.

The prior model probabilities were set equally across model types, as described in Bartoš et al. (2022). In summary, 50% of the total prior model probability was allocated to models assuming the presence of the effect, 50% of the total prior model probability was allocated to models assuming the absence of between-study heterogeneity, and 50% of the total prior model probability was allocated to models assuming the presence of publication bias. Out of the prior model probability allocated to models assuming publication bias, 50% was equally spread across the 6 weight functions, and 50% was equally spread across PET and PEESE models.

Equity, diversity and inclusion statement

The author team consists of both student and senior researchers from multiple disciplines. The study populations encompassed individuals from diverse demographic, socioeconomic, and cultural backgrounds, including participants from low- and middle-income countries.

Patient and public involvement

There was no patient or public involvement in this research.

Results

We successfully extracted 2,239 study-level effect estimates from 215 out of 286 (75.2%) individual meta-analyses included in Singh et al. (2025). The remaining data could not be obtained due to missing study-information tables, forest plots, or supplementary information in the original meta-analyses (see Figure 1 in (Singh et al., 2025), for PRISMA flowchart detailing study search, selection, and extraction). After excluding 306 suspected duplicate effect sizes (13.7%), we retained 1,933 estimates from 201 meta-analyses.

Figure 1 visualizes the data. The first row highlights a notable right skew in funnel plots for each outcome (all Egger tests $p < .0001$; (Egger et al., 1997)). Importantly, funnel plot asymmetry is not necessarily indicative of publication bias (Sterne et al., 2011), especially in the case of meta-meta-analyses where different meta-analyses might concern different interventions and populations. The second row addresses this small-study limitation by visualizing the distribution of z-statistics for each outcome e.g. Bartoš and Schimmack, 2025; Brodeur et al., 2016; Brunner and Schimmack, 2020; van Zwet and Cator, 2021. Publication bias can be identified by sharp discontinuities at the statistical significance criterion and the direction of the effect (highlighted by red dotted vertical lines). While the distribution of z-statistics of memory does not reveal any suspicious patterns, the distribution of z-statistics of general cognition and executive function reveals severe selection for positive effects. The fact that the publication bias seems to predominantly target negative effect sizes explains why publication bias adjustment via 3PSM by Ciria et al. (2023) did not show a noticeable effect size adjustment (3PSM only adjusts for selection on statistical significance).

Below, we report results for each outcome and subgroup separately, and then re-examine the original meta-analyses (see Supplementary Information for detailed results).

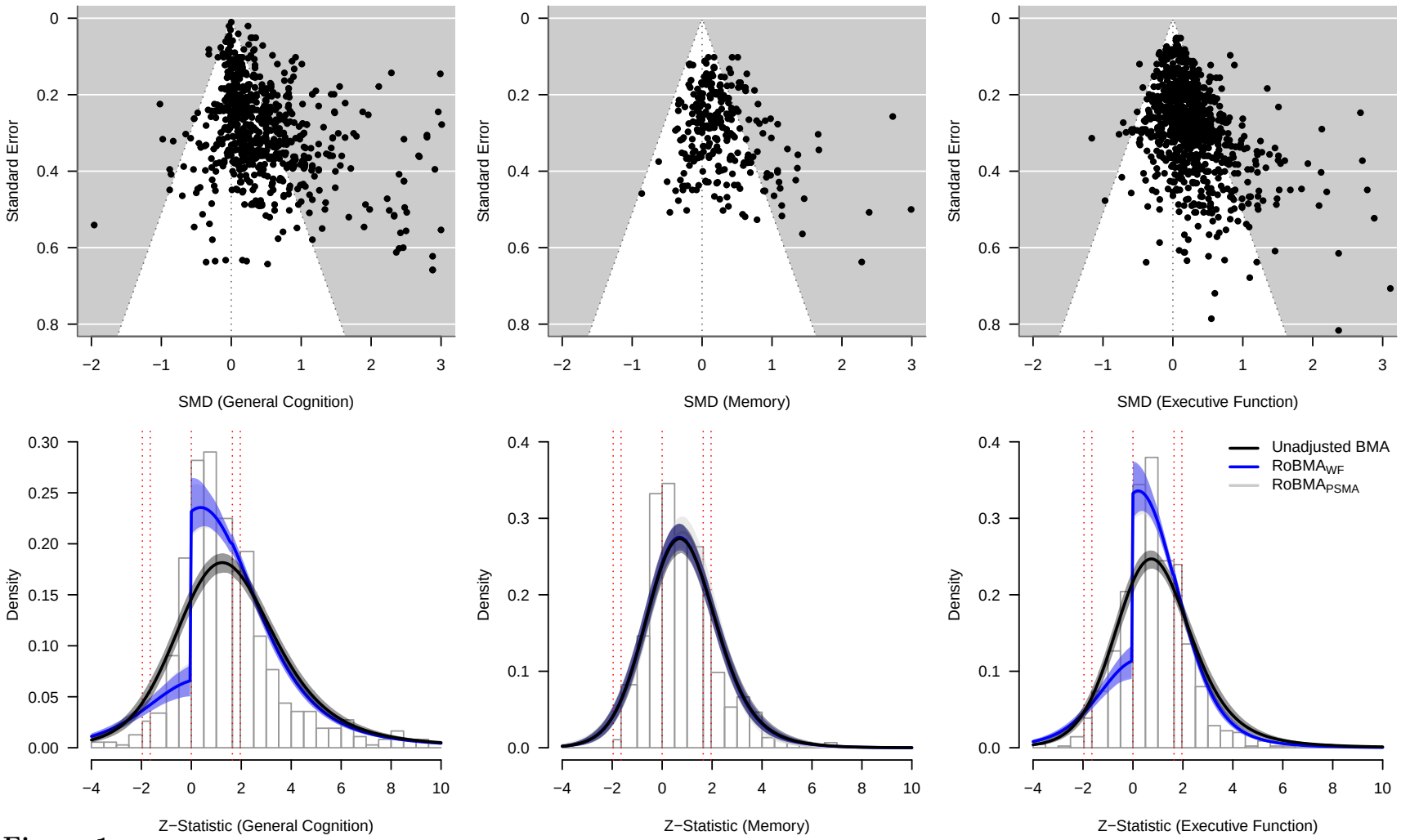


Figure 1

Funnel Plots and Z-Plots of the Effect of Exercise on General Cognition, Memory, and Executive Function

Note: Funnel plots centered at zero indicate funnel plot asymmetry of the study-level effect size estimates in each outcome (12, 0, and 10 effect sizes are out of the plotting range). Z-plots depict histograms of the study-level z-statistics with red dotted vertical lines highlighting selection (discontinuities) for positive, marginally significant, and significant results (13, 1, and 4 z-statistics are out of the plotting range). Density estimates show the marginal posterior predictive distributions of z-statistics under the publication bias-unadjusted BMA model (black), weightfunction-only RoBMA (RoBMA$_{\text{WF}}$, blue) and complete RoBMA model (RoBMA$_{\text{PSMA}}$, grey) which indicated that publication bias-adjusted models fit the data notably better than publication bias unadjusted models (z-plots of RoBMA$_{\text{WF}}$ and RoBMA$_{\text{PSMA}}$ fully coincide for general cognition and executive function). SMD = standardized mean difference.

General cognition

A total of unique 731 effect sizes from 77 (out of 107) meta-analyses of exercise on general cognition were included in the analysis. A three-level publication bias-unadjusted Bayesian model-averaged (BMA) meta-analysis reproduced an overall effect size estimate (SMD = 0.451, 95% CrI $[0.386, 0.517]$) comparable to the publication bias-unadjusted effect size estimate (SMD = 0.42, 95% CI $[0.37, 0.47]$) reported by (Singh et al., 2025). However, the left panel in the second row of Figure 1 highlights that the publication bias-unadjusted BMA meta-analysis (black density) severely misfits the observed distribution of test statistics. The publication bias-adjusted meta-analyses ($\text{RoBMA}_{\text{PSMA}}$ in gray and RoBMA_{WF} in blue density) approximate the observed distribution of z-test statistics to a much better degree. The visualization is further confirmed by extreme evidence of publication bias (Bayes factor in favor of publication bias, $\text{BF}_{\text{bias}} > 1.00 \times 10^6$) regardless of the publication bias-adjustment model (both $\text{RoBMA}_{\text{PSMA}}$ and RoBMA_{WF} produce essentially identical results, i.e., no small-study adjustment; we report only $\text{RoBMA}_{\text{PSMA}}$ results). The estimated weight function highlights the severe suppression of statistically non-significant results (relative probability of publishing negative results $\omega_{p>0.50} = 0.285$, 95% CrI $[0.200, 0.385]$).

Importantly, the notably reduced publication bias-adjusted overall estimate (SMD = 0.208, 95% CrI $[0.075, 0.318]$) supported by extreme evidence for the presence of the effect (Bayes factor in favor of effect, $\text{BF}_{10} = 58.9$) is of little value due to the extreme between-study heterogeneity ($\tau = 0.633$, 95% CrI $[0.580, 0.694]$). In other words, the prediction interval (i.e., true effect sizes consistent with the data) of the publication bias-adjusted overall estimate covers effect sizes consistent with both large unfavorable and large beneficial effects (PI$[-1.045, 1.453]$). The moderator analysis did not show evidence of moderation by any of our moderators (population type, population age, exercise intensity, exercise category, or intervention duration), possibly due to variation in between-study heterogeneity across the moderator levels. Therefore, we performed

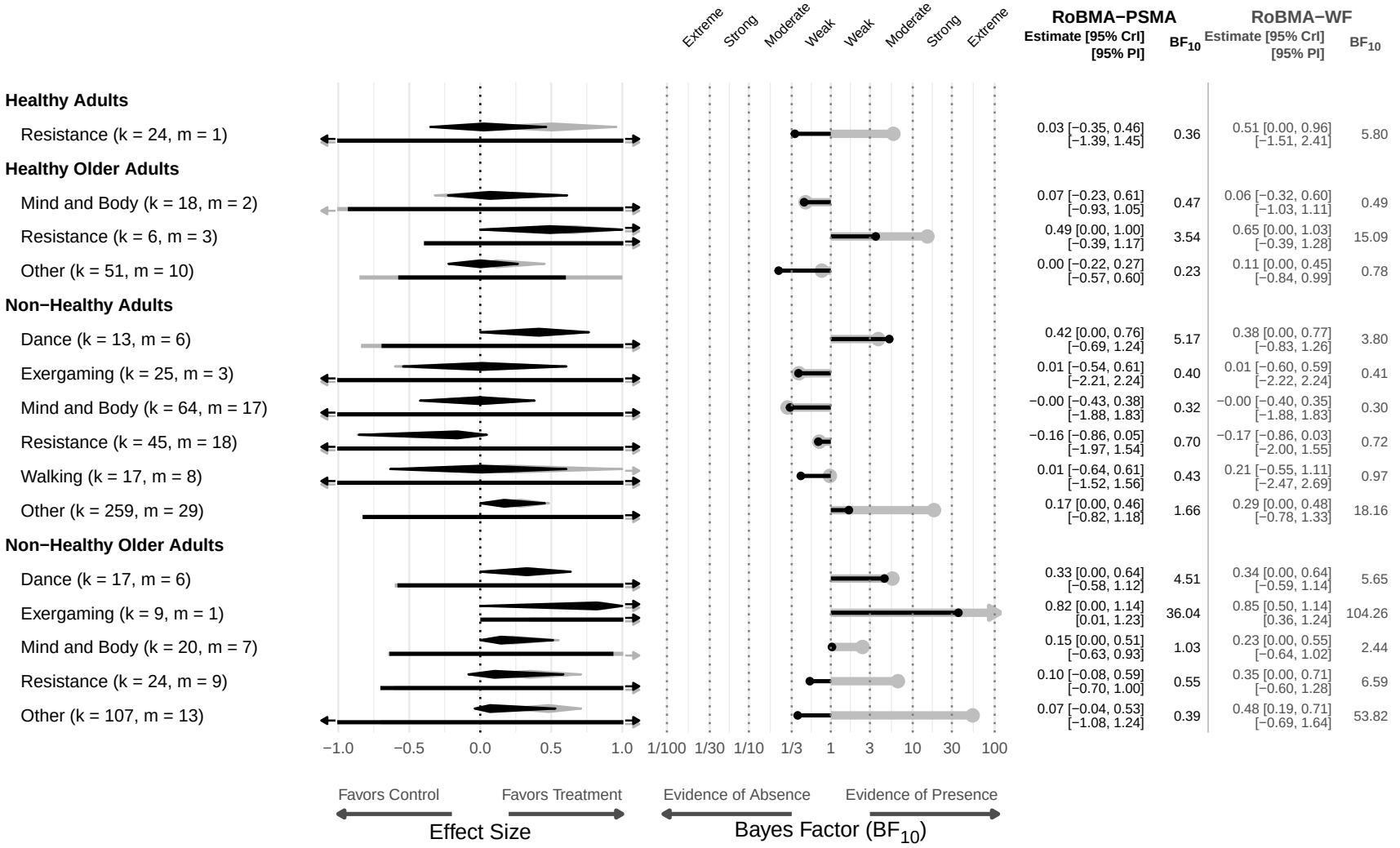


Figure 2

Effects of Exercise on General Cognition Across Populations and Exercise Categories

Note: Diamonds represent the pooled effect size estimates, rectangles represent the prediction interval. $\text{RoBMA}_{\text{PSMA}}$ (black) corresponds to the default settings of RoBMA and RoBMA_{WF} (grey) corresponds to the weight function only version of RoBMA. k = number of estimates, m = the number of meta-analyses, BF_{10} = Bayes factor for the presence of the effect, CrI = credible interval, PI = prediction interval

subgroup analyses by population type, population age, and exercise category.

Figure 2 summarizes the results of subgroup analyses across populations and exercise categories. In healthy adults, the results show an extreme degree of between-study heterogeneity in the effect of resistance training on general cognition with prediction intervals consistent with both large unfavorable and large beneficial effects. In healthy older adults, the effect of resistance training on general cognition seems to be slightly less heterogeneous. The analysis shows a positive pooled effect with mostly positive prediction interval and moderate to strong evidence in favor of the effect. The remaining exercise categories show an extreme degree of between-study heterogeneity and weak to moderate evidence against the pooled effect. In non-healthy adults, the effect of all exercise categories on general cognition is extremely heterogeneous, with moderate and weak to strong evidence for the presence of the pooled effect in the dance and other category. In non-healthy older adults, there seems to be moderate and extreme evidence for the pooled effect of both dance and exergaming on general cognition. For exergaming, the prediction interval is consistent with positive effects only; however, the result is based on only 9 effect size estimates from a single meta-analysis. The remaining exercise categories are extremely heterogeneous, with a considerable discrepancy between $\text{RoBMA}_{\text{PSMA}}$ and RoBMA_{WF} results for the other exercise category due to potential small-study effects.

Memory

A total of unique 301 effect sizes from 39 (out of 62) meta-analyses of exercise on memory were included in the analysis. A three-level publication bias-unadjusted BMA meta-analysis reproduced an overall effect size estimate (SMD = 0.226, 95% CrI $[0.171, 0.284]$) similar to the publication bias-unadjusted effect size estimate (SMD = 0.26, 95% CI $[0.20, 0.32]$) reported by (Singh et al., 2025). The middle panel in the second row of Figure 1 does not show a notable selection for statistically significant or positive results. This is reflected in the inconsistent results of $\text{RoBMA}_{\text{PSMA}}$ and RoBMA_{WF} publication

bias-adjusted results. While $\text{RoBMA}_{\text{PSMA}}$ finds strong evidence for publication bias operating via inflated results in small studies ($\text{BF}_{\text{bias}} = 26.5$; consistent with small-study effects), RoBMA_{WF} finds extreme evidence against publication bias operating via selection for significance or direction of the results ($\text{BF}_{\text{bias}} = 1/44.2$).

The publication bias-adjusted overall estimate under RoBMA_{WF} (SMD = 0.226, 95% CrI $[0.169, 0.282]$) is supported by extreme evidence for the presence of the effect ($\text{BF}_{10} > 1.00 \times 10^6$) and matches the publication bias-unadjusted estimate; however, the publication bias-adjusted overall estimate under $\text{RoBMA}_{\text{PSMA}}$ is shrunken towards zero (SMD = 0.021, 95% CrI $[0.000, 0.217]$) with moderate evidence against the presence of the effect ($\text{BF}_{10} = 1/4.61$). Both models find substantial between-study heterogeneity ($\tau = 0.278$, 95% CrI $[0.233, 0.326]$, $\tau = 0.275$, 95% CrI $[0.230, 0.325]$), resulting in very wide prediction intervals consistent with both medium unfavorable and medium beneficial effects (PI$[-0.324, 0.777]$, PI$[-0.530, 0.578]$; under RoBMA_{WF} and $\text{RoBMA}_{\text{PSMA}}$ respectively). The moderator analysis revealed extreme and moderate evidence for moderation by the exercise category.

Figure 3 summarizes the results of subgroup analyses across populations and exercise categories. In healthy adults, the results show an extreme degree of between-study heterogeneity in the effect of other exercises on memory (prediction intervals are consistent with both large unfavorable and large beneficial effects). In other populations, the results show a substantially lower degree of heterogeneity, which is summarized in the much narrower prediction intervals. In healthy older adults, only the other exercise category shows moderate evidence in favor of the pooled effect. In non-healthy adults, the effect of resistance, walking, and other exercises on memory is consistent with moderate to strong evidence of no benefit. The only exercise category supported by at least weak evidence in favor of the effect, although with substantial between-study heterogeneity, is dance. In non-healthy older adults, the effect of dance on memory is also supported by weak to moderate evidence, accompanied by substantial heterogeneity, and the other exercise

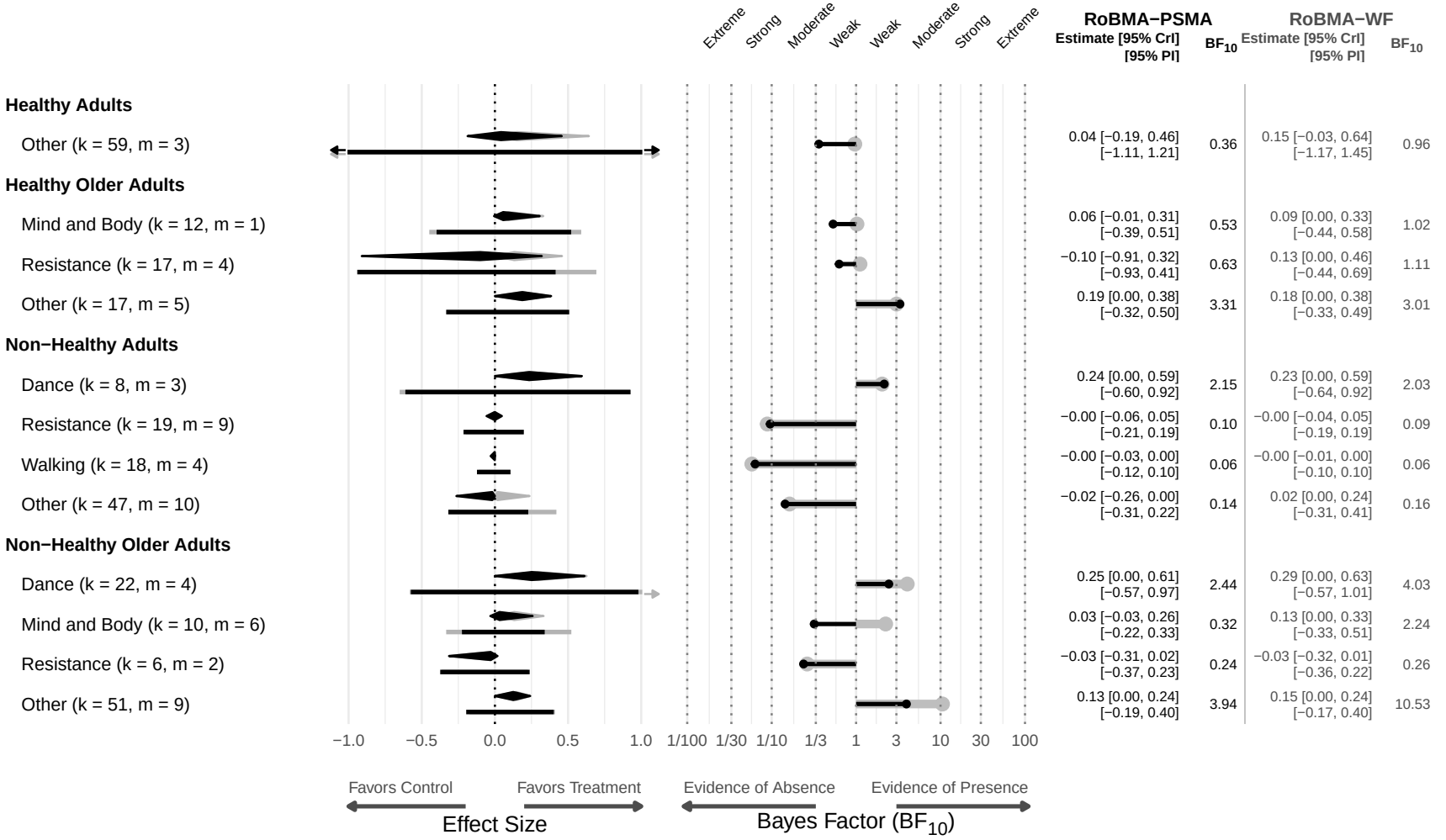


Figure 3

Effects of Exercise on Memory Across Populations and Exercise Categories

Note: Diamonds represent the pooled effect size estimates, rectangles represent the prediction interval. $\text{RoBMA}_{\text{PSMA}}$ (black) corresponds to the default settings of RoBMA and RoBMA_{WF} (grey) corresponds to the weight function only version of RoBMA. k = number of estimates, m = the number of meta-analyses, BF_{10} = Bayes factor for the presence of the effect, CrI = credible interval, PI = prediction interval

category is supported by moderate to strong evidence.

Executive function

A total of 901 effect sizes from 85 (out of 117) meta-analyses of exercise on executive function were included in the analysis. A three-level publication bias-unadjusted BMA meta-analysis reproduced an overall effect size estimate (SMD = 0.245, 95% CrI $[0.199, 0.293]$) consistent with the publication bias-unadjusted effect size estimate (SMD = 0.24, 95% CI $[0.21, 0.27]$) reported by (Singh et al., 2025). Again, the right panel in the second row of Figure 1 highlights that the publication bias-unadjusted meta-analysis (black density) misfits the observed distribution of test statistics. The publication bias-adjusted meta-analysis ($\text{RoBMA}_{\text{PSMA}}$ in gray and RoBMA_{WF} in blue density) approximates the observed distribution of test statistics to a much better degree, which is further confirmed by extreme evidence of publication bias ($\text{BF}_{\text{bias}} > 1.00 \times 10^6$) regardless of the publication bias-adjustment model. $\text{RoBMA}_{\text{PSMA}}$ and RoBMA_{WF} produce essentially identical results (i.e., no small-study adjustment) with the step function highlighting a notable suppression of negative results $\omega_{p>0.50} = 0.338$, 95% CrI $[0.245, 0.442]$.

Since both publication bias models converge to same estimates, we report only $\text{RoBMA}_{\text{PSMA}}$ results. The publication bias-adjusted estimate is reduced to approximately 36% of the original estimate (SMD = 0.088, 95% CrI $[0.000, 0.168]$) supported by moderate evidence for the presence of the effect ($\text{BF}_{10} = 4.79$). Importantly, the analysis highlights substantial between-study heterogeneity ($\tau = 0.370$, 95% CrI $[0.334, 0.411]$), resulting in very wide prediction intervals consistent with both medium unfavorable and large beneficial effects (PI$[-0.649, 0.820]$). The moderator analysis revealed extreme evidence for moderation by the population type.

Figure 4 summarizes the results of subgroup analyses across populations and exercise categories. In healthy children and adolescents, the results show moderate to strong evidence against the effect of any exercise category on executive function, however,

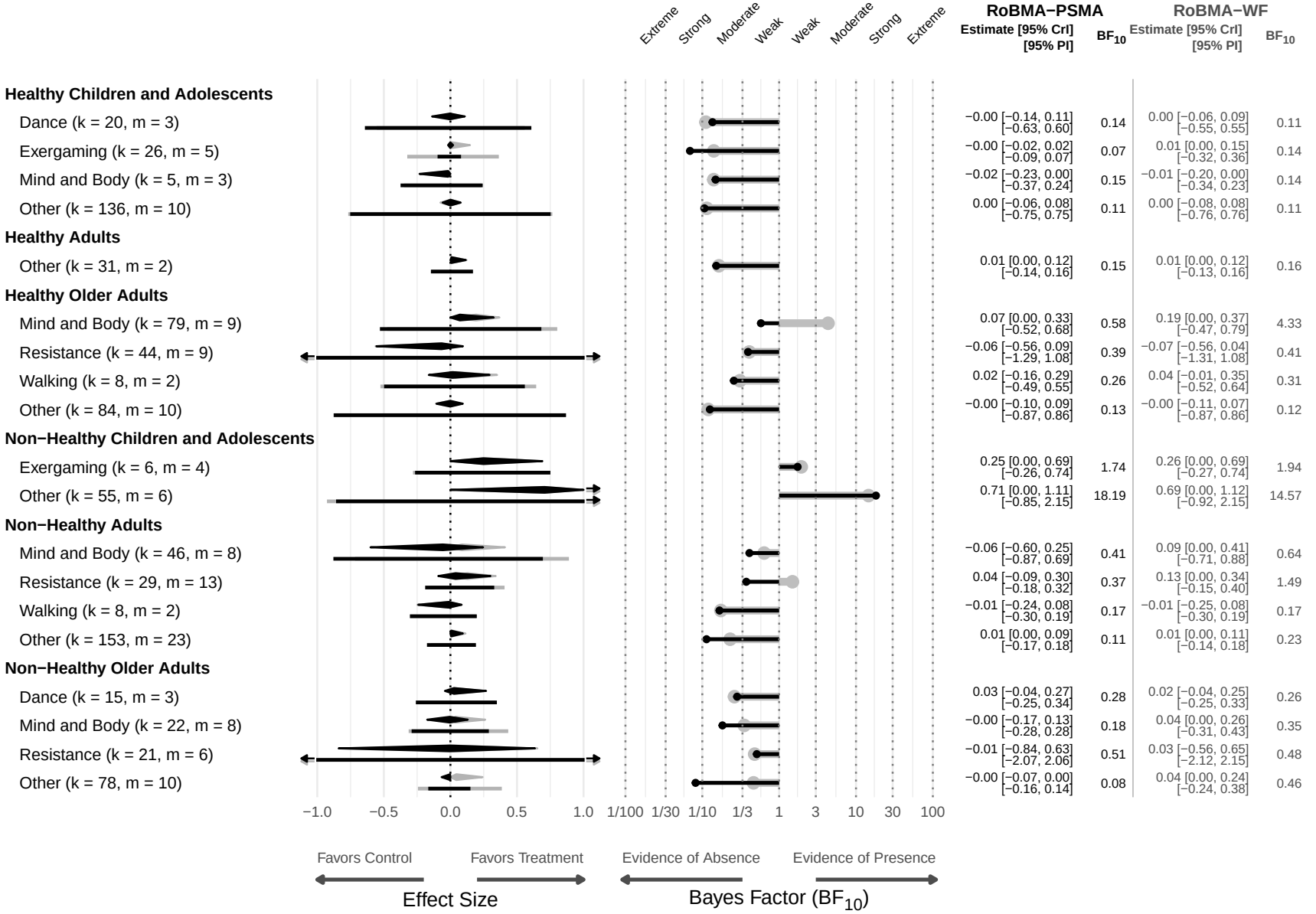


Figure 4

Effects of Exercise on Executive Function Across Populations and Exercise Categories

Note: Diamonds represent the pooled effect size estimates, rectangles represent the prediction interval. $\text{RoBMA}_{\text{PSMA}}$ (black) corresponds to the default settings of RoBMA and RoBMA_{WF} (grey) corresponds to the weight function only version of RoBMA. k = number of estimates, m = the number of meta-analyses, BF_{10} = Bayes factor for the presence of the effect, CrI = credible interval, PI = prediction interval

both the dance and other exercise categories show substantial between-study heterogeneity. In healthy adults, the results show moderate evidence against the effect of other exercise category on executive function with low between-study heterogeneity. In healthy older adults, the effects of exercise categories show substantial between-study heterogeneity with weak to moderate evidence against the effect on executive function, apart from the mind and body exercise category, which shows moderate evidence for the presence of the effect under the RoBMA_{WF} model. In non-healthy children and adolescents, the results show weak evidence for the effect of exergaming on executive function with mostly positive prediction interval. The other exercise category shows strong evidence for the effect; however, it is accompanied by extreme between-study heterogeneity consistent with medium unfavorable and large beneficial effects. In non-healthy adults, there is weak to moderate evidence against the effect of all exercise categories on executive function. The estimates seem to be mostly homogeneous except the mind and body exercise category. In non-healthy older adults, there is weak to strong evidence against the effect of all exercise categories on executive function. The estimates also seem to be homogeneous except for the resistance training exercise category.

Meta-analysis-level re-analysis

To evaluate the impact of publication bias on the previously reported results, we re-examined the individual meta-analyses included in the meta-meta-analysis with the publication bias-unadjusted BMA meta-analysis and both publication bias-adjusted models ($\text{RoBMA}_{\text{PSMA}}$ and RoBMA_{WF}). Figure 5 summarizes the results of individual meta-analyses, each meta-analysis is visualized as a point with size corresponding to the number of estimates.

The first row of Figure 5 visualizes evidence in favor of the presence (or absence) of publication bias in each meta-analysis when analyzed with RoBMA_{WF} and $\text{RoBMA}_{\text{PSMA}}$. We find that each meta-analysis rarely results in strong evidence for either the presence or

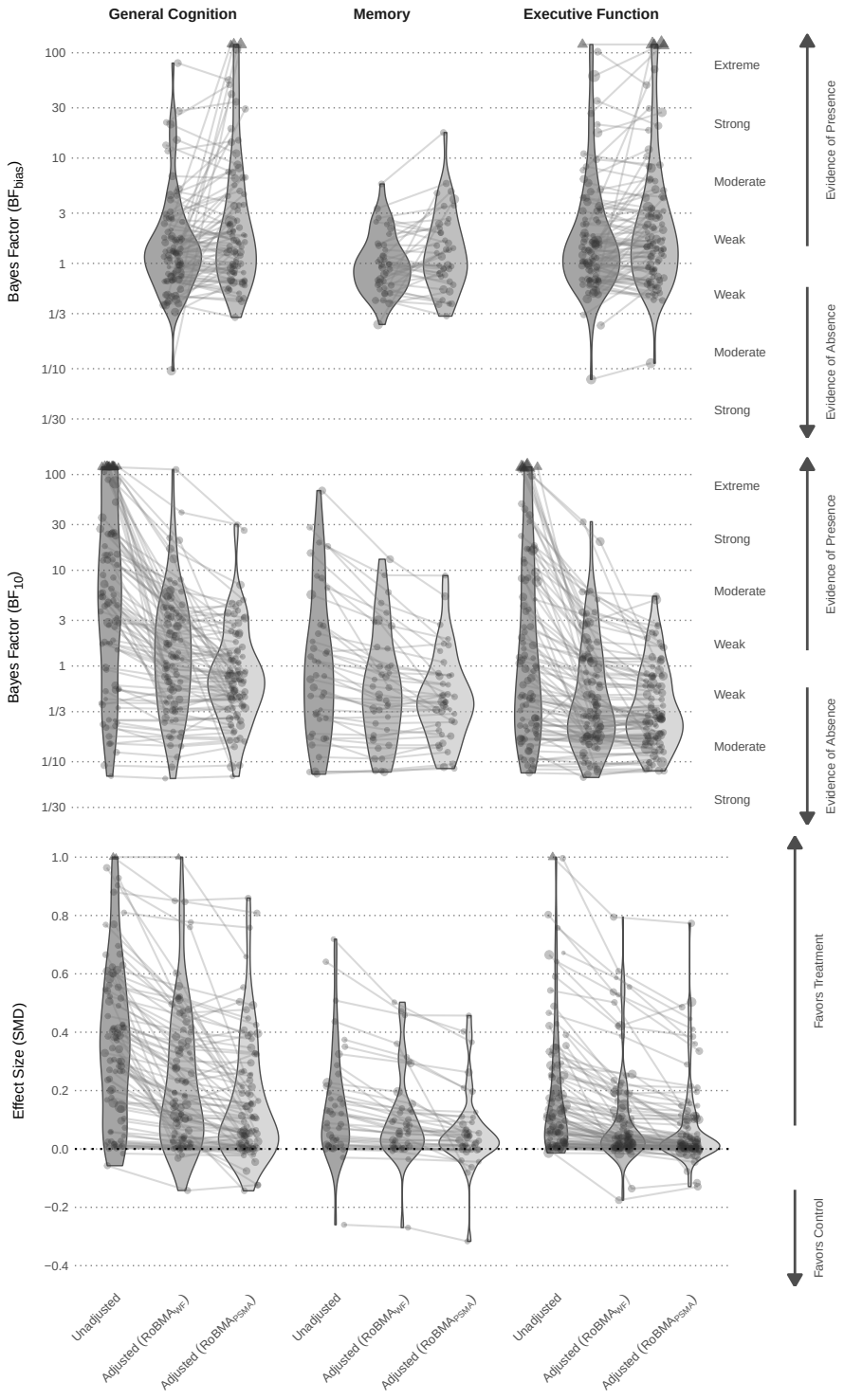


Figure 5

Re-Analysis of the Individual Meta-Analyses

Note: Distribution of Bayes factors for the presence vs. absence of publication bias (first row), effect size (second row), and pooled effect size estimates (third row) for the individual meta-analyses.

absence of publication bias, especially when examining the smaller meta-analyses. Nevertheless, even the more conservative test specifying only weight function publication bias-adjustment RoBMA$_{\text{WF}}$ results in strong evidence for the presence of publication bias ($\text{BF}_{\text{bias}} > 10$) in 5 out of 32 meta-analyses on general cognition, 0 out of 10 meta-analyses on memory, and 6 out of 40 meta-analyses on executive function with at least 10 effect size estimates. Furthermore, RoBMA$_{\text{PSMA}}$ finds strong evidence for the presence of publication bias in 10 out of 32 meta-analyses on general cognition, 0 out of 10 meta-analyses on memory, and 10 out of 40 meta-analyses on executive function. Figures S1, S2, and S3 in the Online Supplements show additional details for each meta-analysis, highlighting that only larger meta-analyses produce sufficient evidence for publication bias.

The second row of Figure 5 visualizes evidence in favor of the presence (or absence) of the pooled effect in each meta-analysis. Interestingly, several meta-analyses find evidence against the effect even prior to any publication bias adjustment: 2 meta-analyses on general cognition, 3 meta-analyses on memory, and 4 meta-analyses on executive function (the first violins). The largest decrease in the evidence for the effect seems to occur already with the weight function only adjustment RoBMA$_{\text{WF}}$, however, RoBMA$_{\text{PSMA}}$ often finds even less evidence for the presence of the effect. While the publication bias-unadjusted BMA meta-analysis finds strong evidence for the presence of the effect ($\text{BF}_{10} > 10$) in 22 out of 32 meta-analyses on general cognition, 2 out of 10 meta-analyses on memory, and 13 out of 40 meta-analyses on executive function with at least 10 effect size estimates, the weight function only publication bias adjustment RoBMA$_{\text{WF}}$ already decreases the number of meta-analyses with strong evidence in favor of the effect to 3 out of 32 meta-analyses on general cognition, 1 out of 10 meta-analyses on memory, and 1 out of 40 meta-analyses on executive function respectively. The further publication bias adjustment with RoBMA$_{\text{PSMA}}$ results in no meta-analysis with at least 10 estimates and strong evidence in favor of the effect. Figures S4, S5, and S6 in the Online Supplements show additional details for each meta-analysis and document the predominant absence of

evidence or the evidence of absence of the pooled effect at the meta-analysis level.

The third row of Figure 5 visualizes the pooled effect size estimates from each meta-analysis. The largest effect size correction occurs in meta-analyses on general cognition, already with the more conservative weight function publication bias adjustment RoBMA_{WF}. Specifically, the distribution of publication bias-unadjusted pooled effect size estimates decreases from the median (interquartile range) $\text{SMD} = 0.36\,(0.14, 0.55)$ to median $\text{SMD} = 0.17\,(0.04, 0.37)$ when applying RoBMA_{WF}, and further to median $\text{SMD} = 0.10\,(0.01, 0.26)$ with $\text{RoBMA}_{\text{PSMA}}$. For meta-analyses on memory, the distribution of pooled effect size estimates decreases from the publication bias-unadjusted median $\text{SMD} = 0.10\,(0.02, 0.22)$ to median $\text{SMD} = 0.06\,(0.01, 0.14)$ when applying RoBMA_{WF}, and further to median $\text{SMD} = 0.03\,(0.00, 0.09)$ with $\text{RoBMA}_{\text{PSMA}}$. Finally, for meta-analyses on executive function, the distribution of pooled effect size estimates decreases from the publication bias-unadjusted median of $\text{SMD} = 0.11\,(0.02, 0.26)$ to median $\text{SMD} = 0.04\,(0.01, 0.15)$ when applying RoBMA_{WF}, and further to median $\text{SMD} = 0.02\,(0.00, 0.10)$ with $\text{RoBMA}_{\text{PSMA}}$. Figures S7, S8, and S9 in the Online Supplements summarize the meta-analysis specific estimates.

Discussion

This study-level meta-meta-analysis examined the effects of exercise on general cognition, memory, and executive function across populations and exercise categories. Overall analysis of 201 meta-analyses with a total of 1,933 effect size estimates revealed substantial publication bias and extreme between-study heterogeneity. The publication bias-adjusted average estimates were notably smaller than previously reported for general cognition and executive function. For memory, the average effect depended on the adjustment specification, remaining positive under the weight-function-only adjustment but shrinking towards zero under the full adjustment. These averages however mask extreme between-study heterogeneity. A fine-grained, publication bias-adjusted analysis by

population and exercise category indicated that only a few interventions show promising evidence for positive effects on cognition. Given the magnitude of publication bias and heterogeneity, strong recommendations that exercise improves cognition are not warranted.

Our results are comparable with previous findings of Ciria et al. (2023), who performed an umbrella review of 24 meta-analyses of exercise on cognition and identified substantial issues in the published literature. Our pooled publication bias-adjusted effect size estimates for general cognition: SMD = 0.208, 95% CrI $[0.075, 0.318]$ and executive function: SMD = 0.088, 95% CrI $[0.000, 0.168]$ are shrunk to a similar degree as the pooled adjusted estimate (SMD = 0.05, 95% CI $[-0.09, 0.14]$) reported by Ciria et al. (2023), which includes an additional adjustment for active controls and baseline imbalances. Our pooled publication bias-adjusted effect size estimate for memory differs between the weight-function only adjustment: SMD = 0.226, 95% CrI $[0.169, 0.282]$ and full adjustment including PET-PEESE SMD = 0.021, 95% CrI $[0.000, 0.217]$ which might be subject to small-study effects in meta-meta-analytic settings. Our results also align with the recent large meta-analysis of acute exercise on cognition in young adults, which found only minimal effects (SMD = 0.13, SD = 0.04) and a comparable degree of extreme between-study heterogeneity (Garrett et al., 2024). Although general cognition intuitively appears to subsume domains such as memory and executive function, in the underlying literature, it is treated as a distinct construct, typically assessed using composite screening measures (e.g., MMSE, MoCA) in older or clinical samples. These measures often yield larger effects than domain-specific tests, reflecting both differences in measurement and selective reporting rather than a contradiction between the results. Importantly, the extreme heterogeneity in the existing literature cannot be ignored; in fact, the results are often consistent with both large unfavorable and large beneficial effects. As such, the pooled effects provide little information on their own and should be interpreted with extreme caution.

The extent of between-study heterogeneity and publication bias cannot be evaluated

when combining meta-analytic summary estimates rather than study-level estimates, as done in Singh et al. (2025). Their meta-analysis-level synthesis provided a valuable overview of the literature but, by design, could not fully capture variability and publication bias among individual trials. Consequently, the pooled effects reported in Singh et al. (2025) (general cognition: SMD = 0.42, 95% CI [0.37, 0.47]; memory: SMD = 0.26, 95% CI [0.20, 0.32]; executive function: SMD = 0.24, 95% CI [0.21, 0.27]) appear larger and more precise than they should be. We therefore suggest re-interpreting the description of "strong evidence ... across all populations" Singh et al., 2025, abstract in light of our finding that the underlying effects are considerably more heterogeneous at the study level.

To disentangle the extreme between-study heterogeneity, we performed population- and exercise category-specific subgroup analyses. While the majority of subgroup analyses resulted in the absence of evidence, several subgroups showed promising evidence in favor of an effect. For general cognition, results indicated benefits of resistance training for healthy older adults and dance and exergaming for non-healthy older adults. For memory, results indicated benefits of dance for non-healthy older adults. For executive function, results indicated benefits of exergaming for non-healthy children and adolescents. In some populations, other exercises that do not fit into any of the major categories showed potential; however, they exhibited highly heterogeneous benefits. Conversely, several subgroup analyses revealed moderate to strong evidence against an effect. For memory, results indicated evidence of no benefit of resistance training for non-healthy adults and non-healthy older adults, as well as evidence of no benefit of walking for non-healthy adults. For executive function, results indicated evidence of no benefit of exergaming and mind and body exercise for healthy children and adolescents, no evidence of benefit from resistance training, evidence of no benefit of walking for non-healthy adults, and evidence of no benefit of mind and body exercise for non-healthy older adults.

Re-examination of the original meta-analyses showed notable publication bias and a severe decrease in evidence favoring the effect, paralleling the decline in pooled effect size

estimates. These results are consistent with previous meta-epidemiological assessments of psychological and medical literature see e.g., Bartoš et al., 2024; Fanelli, 2012; Fanelli et al., 2017. Future meta-analyses should therefore report publication bias-adjusted estimates, a practice currently adopted by only a minority of studies (Ciria et al., 2023; Wu et al., 2025). Furthermore, future meta-meta-analyses should utilize study-level estimates to apply publication bias adjustment and accurately assess between-study heterogeneity. This heterogeneity should be described by either the absolute heterogeneity estimate τ or prediction intervals, as the commonly reported relative metric I^2 often conceals the true degree of between-study heterogeneity (Borenstein et al., 2017; Borenstein, 2024; Rücker et al., 2008).

Despite the scale of the current meta-meta-analysis, several limitations should be acknowledged. First, we extracted study-level estimates as reported in the original meta-analyses; it is therefore possible that we propagated errors and suboptimal choices (such as not computing estimates adjusted for pre-treatment measurement, (Ciria et al., 2023)) present in the existing literature. Second, we were unable to precisely track the origin of all extracted study-level estimates, while we removed 13.7% of estimates due to suspected overlap, it is possible that some estimates could have been included more than once. Third, the observed distribution of test statistics indicated a higher skew near the significance criterion than expected under publication bias alone. This may indicate questionable research practices QRPs, John et al., 2012, which can result in over- or under-adjusted pooled effect size estimates (Bartoš et al., 2022; Irsova et al., 2025; Mathur, 2024). Finally, our investigation was limited to empirical evidence from randomized controlled trials; see Ciria et al. (2023), Dupuy et al. (2024), and Ciria et al. (2024) for detailed discussions on theoretical models, animal models, and neurobiological mechanisms.

Conclusions

In summary, this extensive study-level meta-meta-analysis demonstrates insufficient evidence to recommend exercise unconditionally for cognitive benefits. While we found promising evidence for specific exercise categories and populations (alongside evidence against benefits for others), the substantial heterogeneity precludes any general recommendations. As such, more specific recommendations regarding exercise categories for specific populations are needed with respect to cognitive benefits. Our findings should not dissuade individuals from exercising, as there are many other benefits associated with regular exercise.

Declarations

Ethics approval

Not applicable.

Transparency statement

The manuscript is an honest, accurate, and transparent account of the study. No important aspects of the study have been omitted and any discrepancies from the study as originally planned have been explained.

Data Availability Statement

Code and data are available at Open Science Framework: https://osf.io/egfzt and at https://meta-analysis.cz/exercise/.

Funding

František Bartoš acknowledges support from the Czech Science Foundation (grant no. 23-05227M).

Acknowledgments

Computational resources were provided by the e-INFRA CZ project (ID:90254), supported by the Ministry of Education, Youth and Sports of the Czech Republic.

Conflict of Interest

None.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Gemini for sentence-level editing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Contributions

Conceptualization: FB; Data curation: ML, KB, KH, KK; Formal analysis: FB; Funding acquisition: ZI, TH; Investigation: FB, ML, KB, KH, KK, ZI, TH; Methodology: FB; Project administration: ZI, TH; Resources: ML, KB, KH, KK; Software: FB; Supervision: FB, ZI, TH; Visualization: FB; Writing – original draft: FB; Writing – review & editing: FB, ML, KB, KH, KK, ZI, TH. All authors read and approved the final version of the article.

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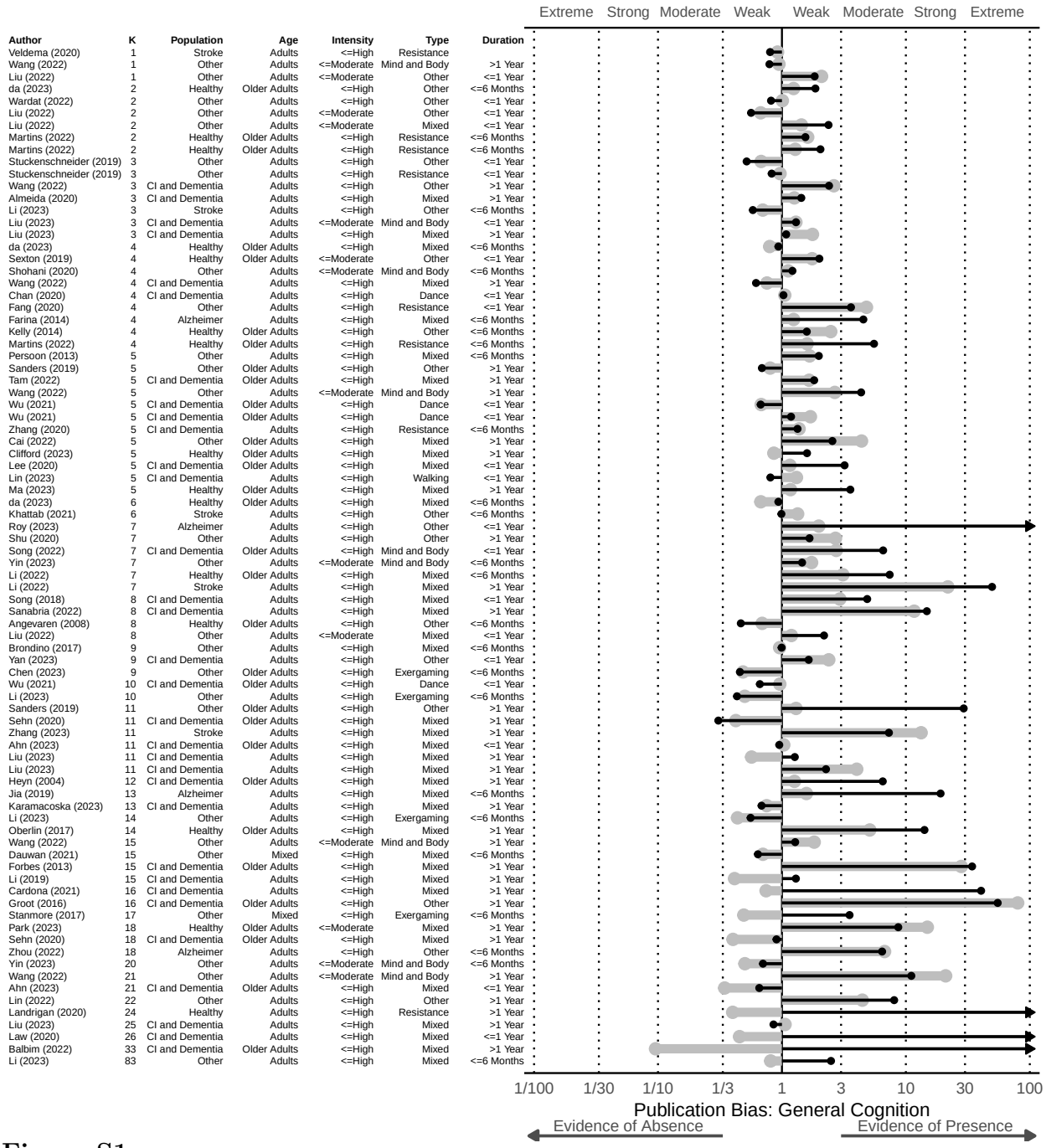
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Online Supplements



Author	K	Population	Age	Intensity	Type	Duration
Veldema (2020)	1	Stroke	Adults	<=High	Resistance	
Wang (2022)	1	Other	Adults	<=Moderate	Mind and Body	>1 Year
Liu (2022)	1	Other	Adults	<=Moderate	Other	<=1 Year
da (2023)	2	Healthy	Older Adults	<=High	Other	<=6 Months
Wardat (2022)	2	Other	Adults	<=High	Other	<=1 Year
Liu (2022)	2	Other	Adults	<=Moderate	Other	<=1 Year
Liu (2022)	2	Other	Adults	<=Moderate	Mixed	<=1 Year
Martins (2022)	2	Healthy	Older Adults	<=High	Resistance	<=6 Months
Martins (2022)	2	Healthy	Older Adults	<=High	Resistance	<=6 Months
Stuckenschneider (2019)	3	Other	Adults	<=High	Other	<=1 Year
Stuckenschneider (2019)	3	Other	Adults	<=High	Resistance	<=1 Year
Wang (2022)	3	CI and Dementia	Adults	<=High	Other	>1 Year
Almeida (2020)	3	CI and Dementia	Adults	<=High	Mixed	>1 Year
Li (2023)	3	Stroke	Adults	<=High	Other	<=6 Months
Liu (2023)	3	CI and Dementia	Adults	<=Moderate	Mind and Body	<=1 Year
Liu (2023)	3	CI and Dementia	Adults	<=High	Mixed	>1 Year
da (2023)	4	Healthy	Older Adults	<=High	Mixed	<=6 Months
Sexton (2019)	4	Healthy	Older Adults	<=Moderate	Other	<=1 Year
Shohani (2020)	4	Other	Adults	<=Moderate	Mind and Body	<=6 Months
Wang (2022)	4	CI and Dementia	Adults	<=High	Mixed	>1 Year
Chan (2020)	4	CI and Dementia	Adults	<=High	Dance	<=1 Year
Fang (2020)	4	Other	Adults	<=High	Resistance	<=1 Year
Farina (2014)	4	Alzheimer	Adults	<=High	Mixed	<=6 Months
Kelly (2014)	4	Healthy	Older Adults	<=High	Other	<=6 Months
Martins (2022)	4	Healthy	Older Adults	<=High	Resistance	<=6 Months
Persoon (2013)	5	Other	Adults	<=High	Mixed	<=6 Months
Sanders (2019)	5	Other	Older Adults	<=High	Other	>1 Year
Tam (2022)	5	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Wang (2022)	5	Other	Adults	<=Moderate	Mind and Body	>1 Year
Wu (2021)	5	CI and Dementia	Older Adults	<=High	Dance	<=1 Year
Wu (2021)	5	CI and Dementia	Older Adults	<=High	Dance	<=1 Year
Zhang (2020)	5	CI and Dementia	Adults	<=High	Resistance	<=6 Months
Cai (2022)	5	Other	Older Adults	<=High	Mixed	>1 Year
Clifford (2023)	5	Healthy	Older Adults	<=High	Mixed	>1 Year
Lee (2020)	5	CI and Dementia	Older Adults	<=High	Mixed	<=1 Year
Lin (2023)	5	CI and Dementia	Adults	<=High	Walking	<=1 Year
Ma (2023)	5	Healthy	Older Adults	<=High	Mixed	>1 Year
da (2023)	6	Healthy	Older Adults	<=High	Mixed	<=6 Months
Khattab (2021)	6	Stroke	Adults	<=High	Other	<=6 Months
Roy (2023)	7	Alzheimer	Adults	<=High	Other	<=1 Year
Shu (2020)	7	Other	Adults	<=High	Other	>1 Year
Song (2022)	7	CI and Dementia	Older Adults	<=High	Mind and Body	<=1 Year
Yin (2023)	7	Other	Adults	<=Moderate	Mind and Body	<=6 Months
Li (2022)	7	Healthy	Older Adults	<=High	Mixed	<=6 Months
Li (2022)	7	Stroke	Adults	<=High	Mixed	>1 Year
Song (2018)	8	CI and Dementia	Adults	<=High	Mixed	<=1 Year
Sanabria (2022)	8	CI and Dementia	Adults	<=High	Mixed	>1 Year
Angevaren (2008)	8	Healthy	Older Adults	<=High	Other	<=6 Months
Liu (2022)	8	Other	Adults	<=Moderate	Mixed	<=1 Year
Brondino (2017)	9	Other	Adults	<=High	Mixed	<=6 Months
Yan (2023)	9	CI and Dementia	Adults	<=High	Other	<=1 Year
Chen (2023)	9	Other	Older Adults	<=High	Exergaming	<=6 Months
Wu (2021)	10	CI and Dementia	Older Adults	<=High	Dance	<=1 Year
Li (2023)	10	Other	Adults	<=High	Exergaming	<=6 Months
Sanders (2019)	11	Other	Older Adults	<=High	Other	>1 Year
Sehn (2020)	11	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Zhang (2023)	11	Stroke	Adults	<=High	Mixed	>1 Year
Ahn (2023)	11	CI and Dementia	Older Adults	<=High	Mixed	<=1 Year
Liu (2023)	11	CI and Dementia	Adults	<=High	Mixed	>1 Year
Liu (2023)	11	CI and Dementia	Adults	<=High	Mixed	>1 Year
Heyn (2004)	12	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Jia (2019)	13	Alzheimer	Adults	<=High	Mixed	<=6 Months
Karamacoska (2023)	13	CI and Dementia	Adults	<=High	Mixed	>1 Year
Li (2023)	14	Other	Adults	<=High	Exergaming	<=6 Months
Oberlin (2017)	14	Healthy	Older Adults	<=High	Mixed	>1 Year
Wang (2022)	15	Other	Adults	<=Moderate	Mind and Body	>1 Year
Dauwan (2021)	15	Other	Mixed	<=High	Mixed	<=6 Months
Forbes (2013)	15	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Li (2019)	15	CI and Dementia	Adults	<=High	Mixed	>1 Year
Cardona (2021)	16	CI and Dementia	Adults	<=High	Mixed	>1 Year
Groot (2016)	16	CI and Dementia	Older Adults	<=High	Other	>1 Year
Stanmore (2017)	17	Other	Mixed	<=High	Exergaming	<=6 Months
Park (2023)	18	Healthy	Older Adults	<=Moderate	Mixed	>1 Year
Sehn (2020)	18	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Zhou (2022)	18	Alzheimer	Adults	<=High	Other	<=6 Months
Yin (2023)	20	Other	Adults	<=Moderate	Mind and Body	<=6 Months
Wang (2022)	21	Other	Adults	<=Moderate	Mind and Body	>1 Year
Ahn (2023)	21	CI and Dementia	Older Adults	<=High	Mixed	<=1 Year
Lin (2022)	22	Other	Adults	<=High	Other	>1 Year
Landrigan (2020)	24	Healthy	Adults	<=High	Resistance	>1 Year
Liu (2023)	25	CI and Dementia	Adults	<=High	Mixed	>1 Year
Law (2020)	26	CI and Dementia	Adults	<=High	Mixed	<=1 Year
Balbim (2022)	33	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Li (2023)	83	Other	Adults	<=High	Mixed	<=6 Months

Figure S1

Evidence in Favor or Against Publication Bias for Each Meta-Analysis on the Effect of Exercise on General Cognition

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.

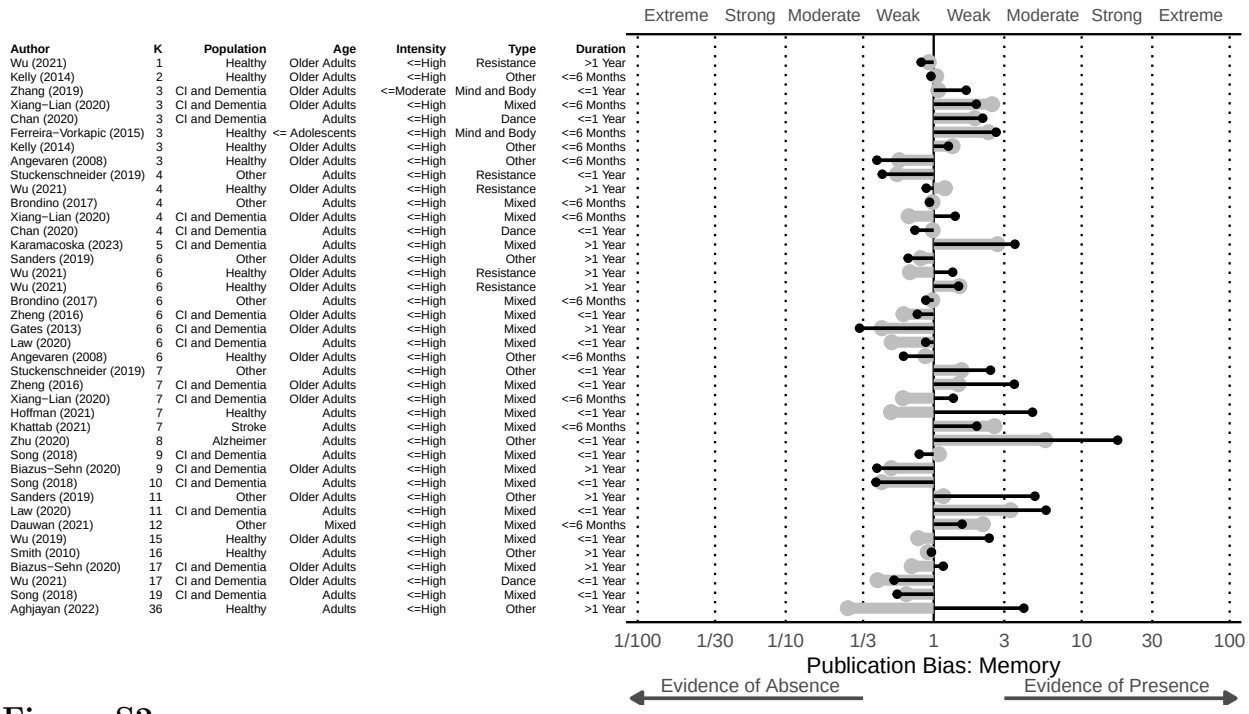


Figure S2

Evidence in Favor or Against Publication Bias for Each Meta-Analysis on the Effect of Exercise on Memory

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.

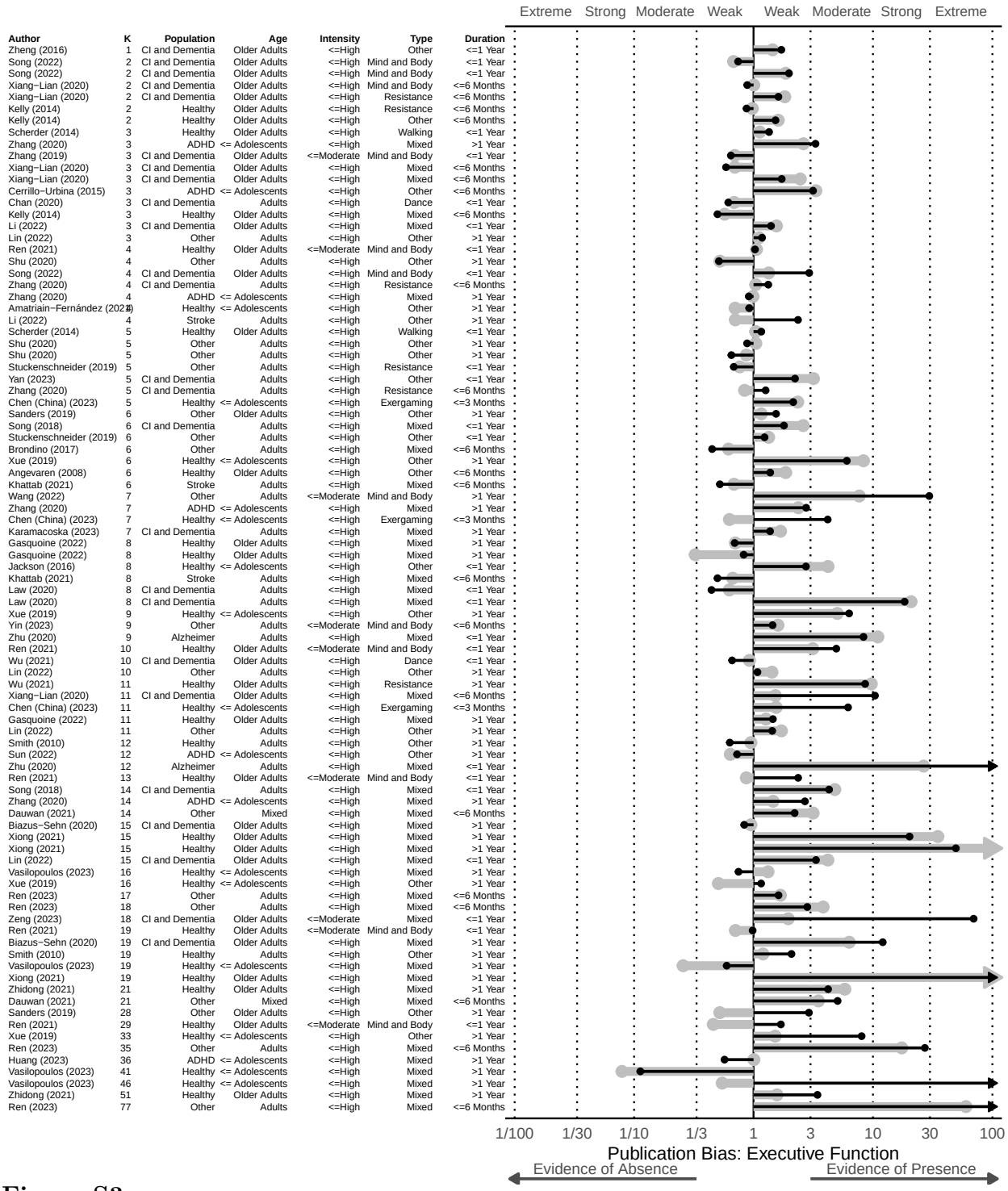


Figure S3

Evidence in Favor or Against Publication Bias for Each Meta-Analysis on the Effect of Exercise on Executive Function

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.

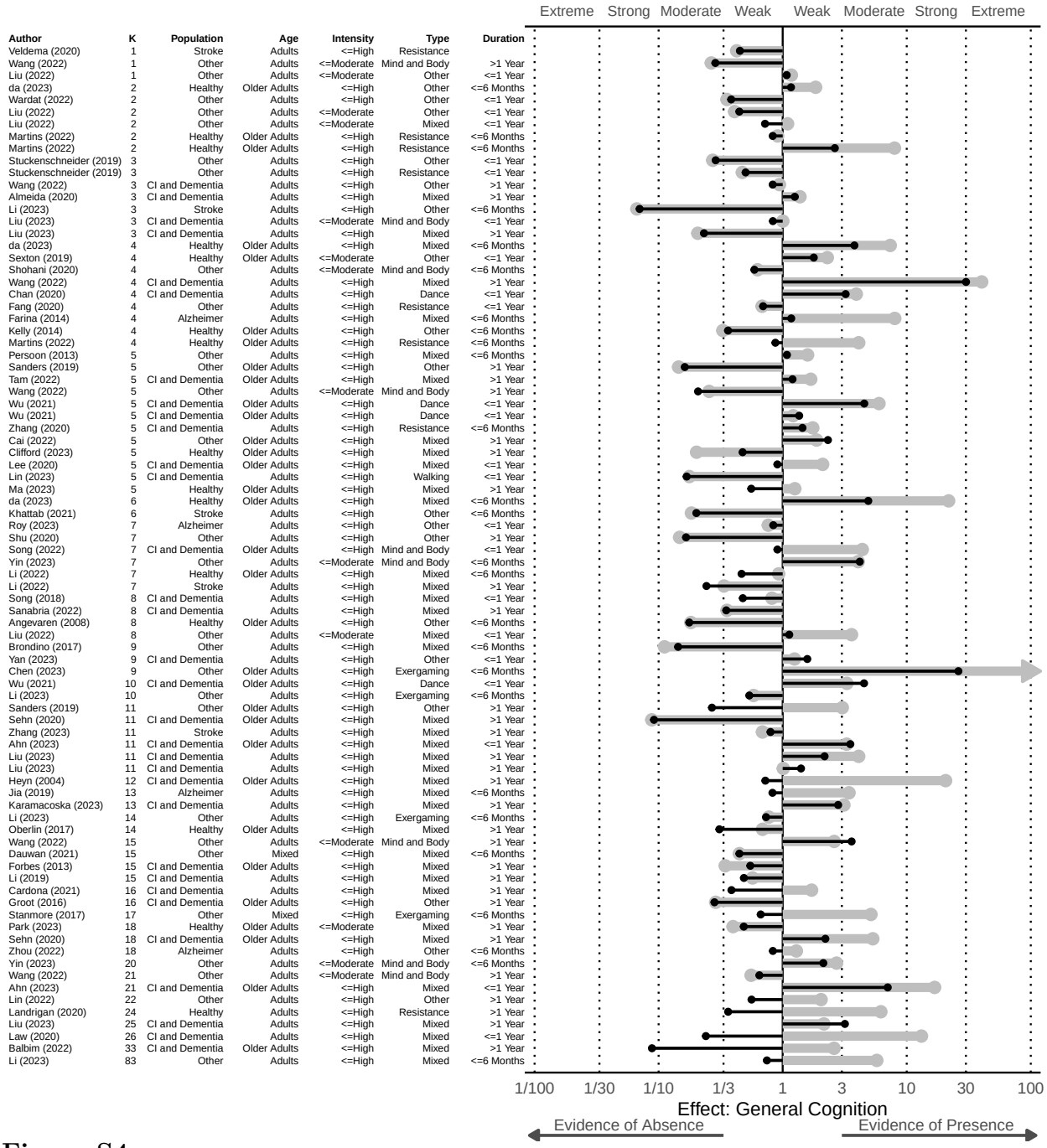


Figure S4

Evidence in Favor or Against Effect for Each Meta-Analysis on the Effect of Exercise on General Cognition

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.

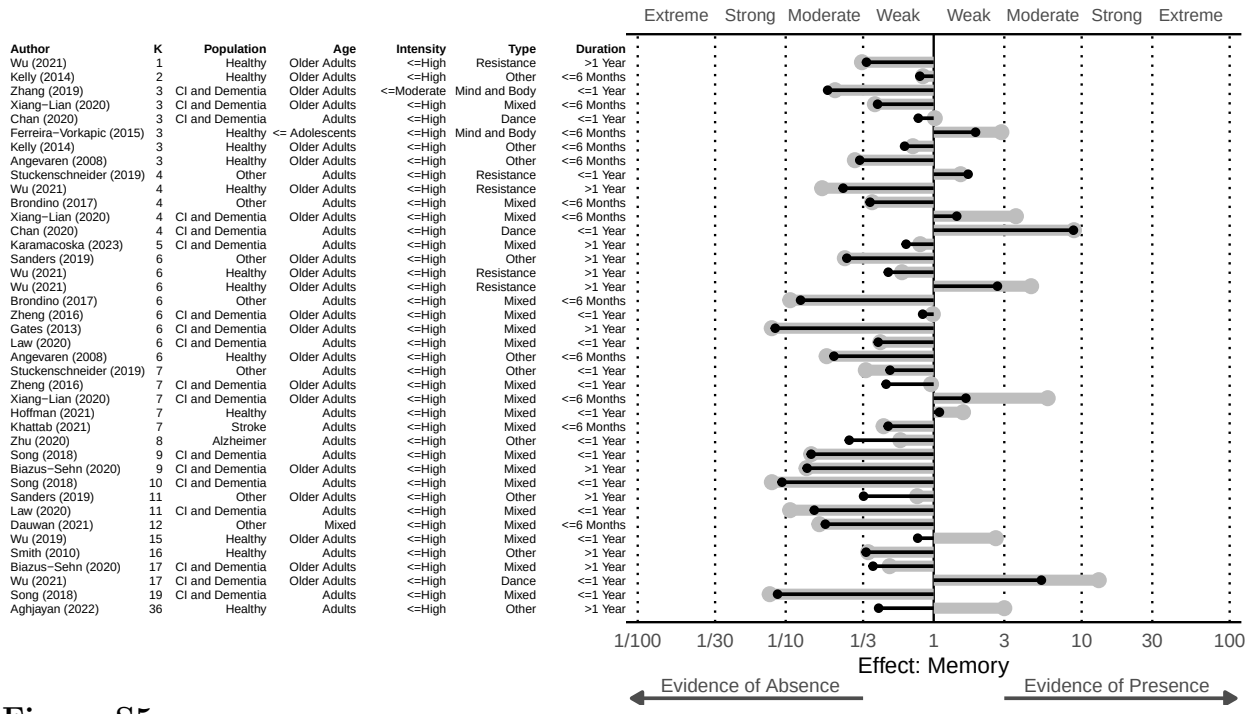
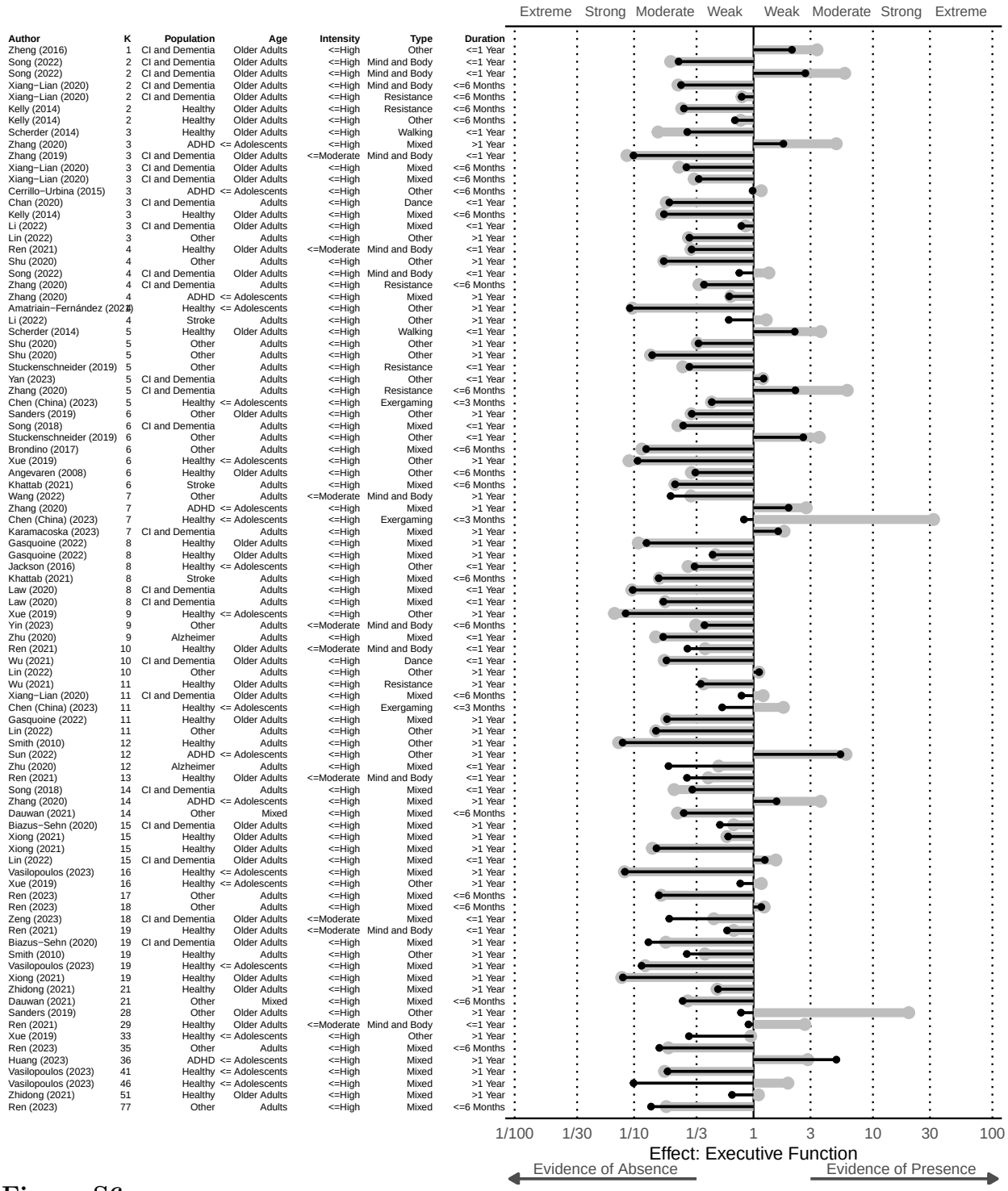


Figure S5

Evidence in Favor or Against Effect for Each Meta-Analysis on the Effect of Exercise on Memory

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.



Author	K	Population	Age	Intensity	Type	Duration
Zheng (2016)	1	CI and Dementia	Older Adults	<=High	Other	<=1 Year
Song (2022)	2	CI and Dementia	Older Adults	<=High	Mind and Body	<=1 Year
Song (2022)	2	CI and Dementia	Older Adults	<=High	Mind and Body	<=1 Year
Xiang−Lian (2020)	2	CI and Dementia	Older Adults	<=High	Mind and Body	<=6 Months
Xiang−Lian (2020)	2	CI and Dementia	Older Adults	<=High	Resistance	<=6 Months
Kelly (2014)	2	Healthy	Older Adults	<=High	Resistance	<=6 Months
Kelly (2014)	2	Healthy	Older Adults	<=High	Other	<=6 Months
Scherder (2014)	3	Healthy	Older Adults	<=High	Walking	<=1 Year
Zhang (2020)	3	ADHD	<= Adolescents	<=High	Mixed	>1 Year
Zhang (2019)	3	CI and Dementia	Older Adults	<=Moderate	Mind and Body	<=1 Year
Xiang−Lian (2020)	3	CI and Dementia	Older Adults	<=High	Mixed	<=6 Months
Xiang−Lian (2020)	3	CI and Dementia	Older Adults	<=High	Mixed	<=6 Months
Cerrillo−Urbina (2015)	3	ADHD	<= Adolescents	<=High	Other	<=6 Months
Chan (2020)	3	CI and Dementia	Adults	<=High	Dance	<=1 Year
Kelly (2014)	3	Healthy	Older Adults	<=High	Mixed	<=6 Months
Li (2022)	3	CI and Dementia	Older Adults	<=High	Mixed	<=1 Year
Lin (2022)	3	Other	Adults	<=High	Other	>1 Year
Ren (2021)	4	Healthy	Older Adults	<=Moderate	Mind and Body	<=1 Year
Shu (2020)	4	Other	Adults	<=High	Other	>1 Year
Song (2022)	4	CI and Dementia	Older Adults	<=High	Mind and Body	<=1 Year
Zhang (2020)	4	CI and Dementia	Adults	<=High	Resistance	<=6 Months
Zhang (2020)	4	ADHD	<= Adolescents	<=High	Mixed	>1 Year
Amatriain−Fernández (2021)	4	Healthy	<= Adolescents	<=High	Other	>1 Year
Li (2022)	4	Stroke	Adults	<=High	Other	>1 Year
Scherder (2014)	5	Healthy	Older Adults	<=High	Walking	<=1 Year
Shu (2020)	5	Other	Adults	<=High	Other	>1 Year
Shu (2020)	5	Other	Adults	<=High	Other	>1 Year
Stuckenschneider (2019)	5	Other	Adults	<=High	Resistance	<=1 Year
Yan (2023)	5	CI and Dementia	Adults	<=High	Other	<=1 Year
Zhang (2020)	5	CI and Dementia	Adults	<=High	Resistance	<=6 Months
Chen (China) (2023)	5	Healthy	<= Adolescents	<=High	Exergaming	<=3 Months
Sanders (2019)	6	Other	Older Adults	<=High	Other	>1 Year
Song (2018)	6	CI and Dementia	Adults	<=High	Mixed	<=1 Year
Stuckenschneider (2019)	6	Other	Adults	<=High	Other	<=1 Year
Brondino (2017)	6	Other	Adults	<=High	Mixed	<=6 Months
Xue (2019)	6	Healthy	<= Adolescents	<=High	Other	>1 Year
Angevaren (2008)	6	Healthy	Older Adults	<=High	Other	<=6 Months
Khattab (2021)	6	Stroke	Adults	<=High	Mixed	<=6 Months
Wang (2022)	7	Other	Adults	<=Moderate	Mind and Body	>1 Year
Zhang (2020)	7	ADHD	<= Adolescents	<=High	Mixed	>1 Year
Chen (China) (2023)	7	Healthy	<= Adolescents	<=High	Exergaming	<=3 Months
Karamacoska (2023)	7	CI and Dementia	Adults	<=High	Mixed	>1 Year
Gasquoine (2022)	8	Healthy	Older Adults	<=High	Mixed	>1 Year
Gasquoine (2022)	8	Healthy	Older Adults	<=High	Mixed	>1 Year
Jackson (2016)	8	Healthy	<= Adolescents	<=High	Other	<=1 Year
Khattab (2021)	8	Stroke	Adults	<=High	Mixed	<=6 Months
Law (2020)	8	CI and Dementia	Adults	<=High	Mixed	<=1 Year
Law (2020)	8	CI and Dementia	Adults	<=High	Mixed	<=1 Year
Xue (2019)	9	Healthy	<= Adolescents	<=High	Other	>1 Year
Yin (2023)	9	Other	Adults	<=Moderate	Mind and Body	<=6 Months
Zhu (2020)	9	Alzheimer	Adults	<=High	Mixed	<=1 Year
Ren (2021)	10	Healthy	Older Adults	<=Moderate	Mind and Body	<=1 Year
Wu (2021)	10	CI and Dementia	Older Adults	<=High	Dance	<=1 Year
Lin (2022)	10	Other	Adults	<=High	Other	>1 Year
Wu (2021)	11	Healthy	Older Adults	<=High	Resistance	>1 Year
Xiang−Lian (2020)	11	CI and Dementia	Older Adults	<=High	Mixed	<=6 Months
Chen (China) (2023)	11	Healthy	<= Adolescents	<=High	Exergaming	<=3 Months
Gasquoine (2022)	11	Healthy	Older Adults	<=High	Mixed	>1 Year
Lin (2022)	11	Other	Adults	<=High	Other	>1 Year
Smith (2010)	12	Healthy	Adults	<=High	Other	>1 Year
Sun (2022)	12	ADHD	<= Adolescents	<=High	Other	>1 Year
Zhu (2020)	12	Alzheimer	Adults	<=High	Mixed	<=1 Year
Ren (2021)	13	Healthy	Older Adults	<=Moderate	Mind and Body	<=1 Year
Song (2018)	14	CI and Dementia	Adults	<=High	Mixed	<=1 Year
Zhang (2020)	14	ADHD	<= Adolescents	<=High	Mixed	>1 Year
Dauwan (2021)	14	Other	Mixed	<=High	Mixed	<=6 Months
Biazus−Sehn (2020)	15	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Xiong (2021)	15	Healthy	Older Adults	<=High	Mixed	>1 Year
Xiong (2021)	15	Healthy	Older Adults	<=High	Mixed	>1 Year
Lin (2022)	15	CI and Dementia	Older Adults	<=High	Mixed	<=1 Year
Vasilopoulos (2023)	16	Healthy	<= Adolescents	<=High	Mixed	>1 Year
Xue (2019)	16	Healthy	<= Adolescents	<=High	Other	>1 Year
Ren (2023)	17	Other	Adults	<=High	Mixed	<=6 Months
Ren (2023)	18	Other	Adults	<=High	Mixed	<=6 Months
Zeng (2023)	18	CI and Dementia	Older Adults	<=Moderate	Mixed	<=1 Year
Ren (2021)	19	Healthy	Older Adults	<=Moderate	Mind and Body	<=1 Year
Biazus−Sehn (2020)	19	CI and Dementia	Older Adults	<=High	Mixed	>1 Year
Smith (2010)	19	Healthy	Adults	<=High	Other	>1 Year
Vasilopoulos (2023)	19	Healthy	<= Adolescents	<=High	Mixed	>1 Year
Xiong (2021)	19	Healthy	Older Adults	<=High	Mixed	>1 Year
Zhidong (2021)	21	Healthy	Older Adults	<=High	Mixed	>1 Year
Dauwan (2021)	21	Other	Mixed	<=High	Mixed	<=6 Months
Sanders (2019)	28	Other	Older Adults	<=High	Other	>1 Year
Ren (2021)	29	Healthy	Older Adults	<=Moderate	Mind and Body	<=1 Year
Xue (2019)	33	Healthy	<= Adolescents	<=High	Other	>1 Year
Ren (2023)	35	Other	Adults	<=High	Mixed	<=6 Months
Huang (2023)	36	ADHD	<= Adolescents	<=High	Mixed	>1 Year
Vasilopoulos (2023)	41	Healthy	<= Adolescents	<=High	Mixed	>1 Year
Vasilopoulos (2023)	46	Healthy	<= Adolescents	<=High	Mixed	>1 Year
Zhidong (2021)	51	Healthy	Older Adults	<=High	Mixed	>1 Year
Ren (2023)	77	Other	Adults	<=High	Mixed	<=6 Months

Figure S6

Evidence in Favor or Against Effect for Each Meta-Analysis on the Effect of Exercise on Executive Function

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.

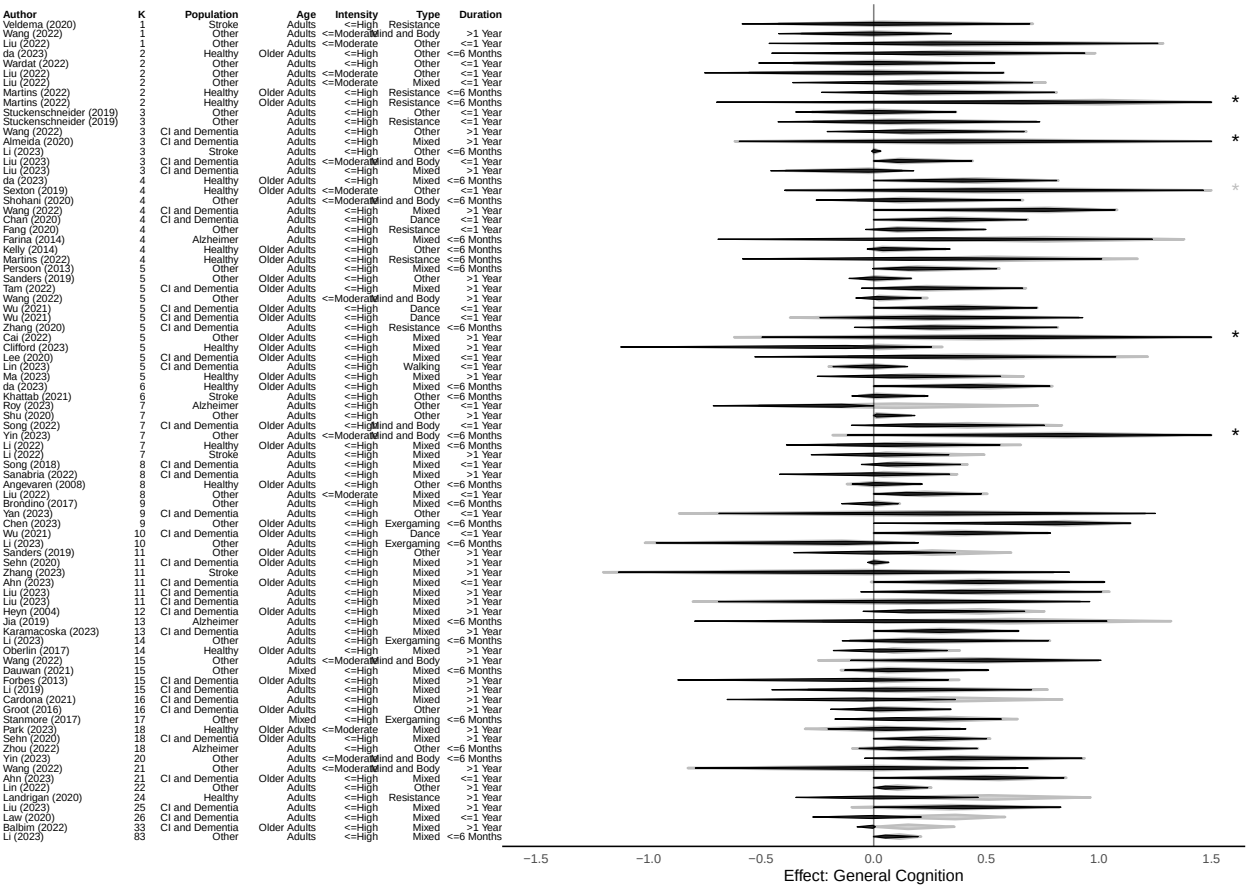


Figure S7

Pooled Effect Size Estimate for Each Meta-Analysis on the Effect of Exercise on General Cognition

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.

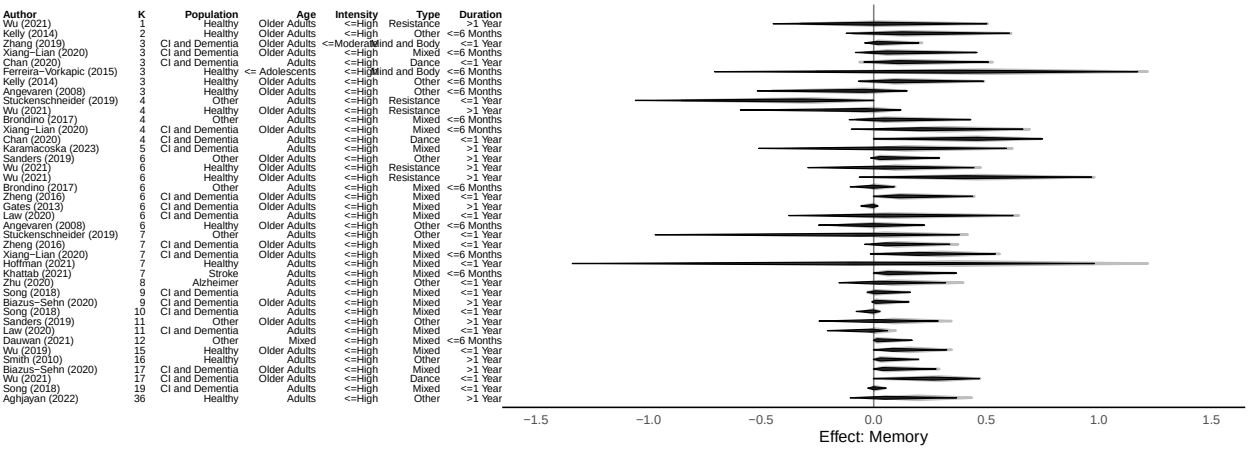


Figure S8

Pooled Effect Size Estimate for Each Meta-Analysis on the Effect of Exercise on Memory

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.

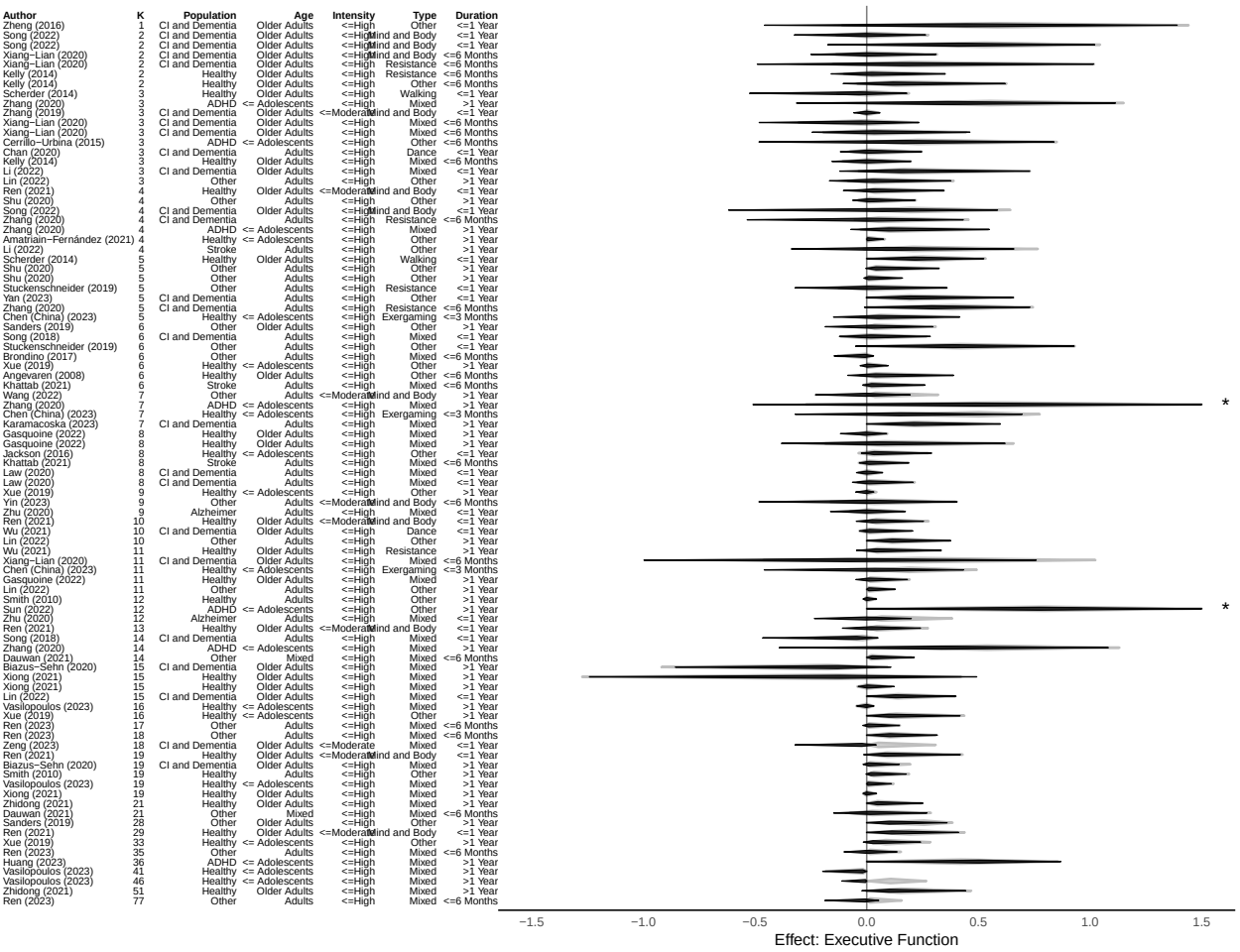


Figure S9

Pooled Effect Size Estimate for Each Meta-Analysis on the Effect of Exercise on Executive Function

Note: $\text{RoBMA}_{\text{PSMA}}$ results in black, RoBMA_{WF} results in grey.